VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY
FACULTY OF COMPUTER SCIENCE AND ENGINEERING

ASSIGNMENT 1
EXPLORATORY DATA ANALYSIS

Programming for Artificial Intelligence and Data Science (CO3135)

Lecturer: Lê Thành Sách

No.	Student Name	Student ID
1	Phan Trần Ý Nhi	2252583
2	Nguyễn Thanh Bình	2252082
3	Lê Mạnh Hưng	2352428
4	Nguyễn Anh Minh Tuệ	2353281

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Team Member Responsibilitie

No.	Student ID	Full Name	Assigned Tasks	Contribution
1	2252583	Phan Trần Ý Nhi	Multimodal	100%
2	2252082	Nguyễn Thanh Bình	Image	100%
3	2352428	Lê Mạnh Hưng	Tabular	100%
4	2353281	Nguyễn Anh Minh Tuệ	Text	100%

Landpage: https://nhiwentwest.github.io/breakthrough-p4ai-ds/

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1 Introduction

The rapid growth of multimodal data in real-world applications has made **Exploratory Data Analysis (EDA)** a critical first step before model development. In many practical pipelines, data are no longer purely numerical: modern systems must process structured records, natural language, visual content, and cross-modal relationships simultaneously. Consequently, robust EDA should not only summarize each modality in isolation, but also evaluate their consistency when modalities interact.

In this project, we conduct a unified EDA across four complementary views: **Tabular**, **Text**, **Image**, and **Multimodal** data. The objective is to build a reliable understanding of data quality, distributional characteristics, and potential risk factors that may affect downstream learning and evaluation.

Tabular EDA provides global dataset diagnostics through descriptive statistics, missing-value patterns, class balance, and feature-level variation. This view is essential for detecting structural issues (e.g., skewed distributions, sparse fields, or unstable ranges) that can distort training behavior.

Text EDA examines lexical and semantic properties of captions, including length distribution, vocabulary size, frequent unigrams/bigrams, and coverage of key semantic cues (e.g., color and spatial terms). These statistics reveal annotation style, linguistic diversity, and potential limitations in language supervision.

Image EDA characterizes visual signals directly from pixel content, such as brightness, texture, and RGB-channel dominance across categories. This analysis helps quantify how visually separable classes are and whether low-level visual priors align with semantic descriptions.

Multimodal EDA then bridges image and text by measuring caption agreement within each image bundle, contradiction patterns between textual claims and visual priors, and uncertainty indicators for noisy samples. Rather than relying on simple keyword overlap, our approach uses an image-level diagnostic framework to produce interpretable noise scores and confidence levels.

Overall, this report aims to answer three practical questions before training multimodal retrieval models: (1) *Is the dataset internally consistent and statistically stable?* (2) *Do text and image modalities provide complementary and aligned supervision?* (3) *Where are the likely failure points (noise, bias, or drift) that require mitigation?*

By integrating evidence from all four modalities, the analysis provides a principled foundation for model design, data cleaning, and evaluation planning.

2 EDA

2.1 Text

2.1.1 Dataset Overview: Twitter Financial News Sentiment

The *Twitter Financial News Sentiment* dataset is an English-language corpus for financial sentiment classification. It contains short finance-related tweets covering market updates, analyst actions, company announcements, and stock-specific news, and is intended for multi-class text classification in a real-world social media setting.

The dataset was collected using the Twitter API and currently supports the **multi-class classification** task for **sentiment analysis**. In total, it contains **11,932** annotated documents (we only use 9938 training datasets) with three sentiment labels:

- `LABEL_0`: Bearish
- `LABEL_1`: Bullish
- `LABEL_2`: Neutral

Each record includes two primary fields: `text` (tweet content) and `label` (sentiment class). Because the source is Twitter, the data often includes ticker symbols (e.g., `$AAPL`), URLs, and concise headline-style writing, which makes it suitable for both exploratory data analysis and practical NLP modeling in finance.

Data Splits The dataset is provided in two predefined splits:

Dataset Split	Number of Instances
Train	9,938
Validation	2,486

2.1.2 Data Cleaning and Missing Data Analysis

Before model training, the dataset is checked for null values to ensure structural consistency and improve downstream stability. The pipeline computes

missingness per feature and summarizes both counts and percentages. If missing values are detected, their distribution is visualized with a Plotly bar chart so that problematic columns can be identified quickly.

After this inspection stage, incomplete rows are removed from the dataset. This step prevents runtime issues during vectorization and avoids introducing noise from partially observed samples. In this dataset, the key analytical fields are `text` and `label`; therefore, dropping rows with null entries ensures that every training example has valid text content and a valid sentiment target.

Why this step matters Handling missingness early improves reproducibility and model reliability. It ensures that tokenization, feature extraction, and classification are performed on complete observations only, reducing bias and preventing unstable behavior in later EDA and machine learning stages.

To prepare the Twitter corpus for downstream sentiment modeling, a lightweight rule-based preprocessing pipeline is applied. The implementation uses three core functions: `remove_urls`, `handle_company`, and `clean_text`. Together, these steps reduce noise while preserving sentiment-relevant linguistic content.

URL Removal The `remove_urls(text)` function removes hyperlinks using the regular expression `https?://\S+|www\S+`. This strips tokens beginning with `http`, `https`, or `www`, which are typically non-informative for sentiment classification and can otherwise increase vocabulary sparsity.

Company Symbol Handling The `handle_company(text)` function replaces cashtags (e.g., `$TSLA`, `$AAPL`) with a placeholder token `<code>`. This standardization prevents the model from overfitting to specific ticker strings while still preserving the structural signal that a company reference existed in the original tweet.

Text Cleaning The `clean_text(text)` function performs final normalization in three stages:

- Removes the temporary `<code>` placeholder.
- Removes non-alphabetic characters using $[^a - zA - Z\backslash s]$.
- `Collapses multiple spaces into a single space and trims leading/trailing whitespace.`

After this pipeline, each tweet is transformed into cleaner alphabetical text with reduced noise and improved consistency, which supports more stable tokenization, feature extraction, and sentiment learning.

As part of the initial data quality control, the dataset was evaluated for duplicate entries and missing values to ensure the integrity of downstream analysis.

Duplicate Handling The analysis identified **153 duplicate records**, corresponding to approximately **1.60%** of the initial dataset. These redundant rows were removed to avoid frequency bias during model training. After deduplication, the working dataset contained **9,390 rows** and **3 columns**.

Missing Values Check After removing duplicates, the dataset was scanned for null values across all features. The check confirmed that the dataset was fully complete, with **zero missing values**. Therefore, no imputation and no additional row removal were required.

Transition to the Next Step With duplicates removed and completeness verified, the dataset passes the core data-quality gate and proceeds to the next preprocessing stage: feature extraction and text normalization.

listings xcolor

2.1.3 Category Distribution

The pie chart reveals a clear and substantial **class imbalance** in the sentiment labels. Specifically, **Label 2** is the dominant class, accounting for approximately **64.6%** of all samples. In contrast, **Label 1** represents about **20.2%**, while **Label 0** is the smallest group at roughly **15.2%**.

This large gap indicates that the dataset is strongly skewed toward neutral news. If the raw distribution is used directly for training, a model may become biased toward predicting Label 2 in order to maximize overall accuracy. As a result, performance on minority classes can degrade, especially in terms of recall and class-wise F1-score. Therefore, this imbalance should be explicitly addressed in later modeling stages through strategies such as class weighting, resampling, or threshold tuning.

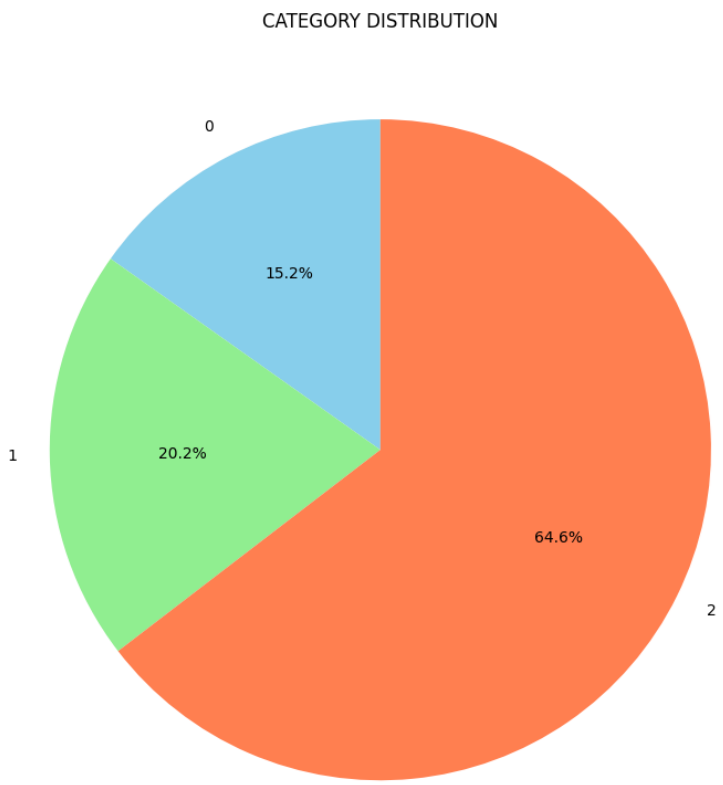


Figure 1: Category distribution of sentiment labels in the dataset.

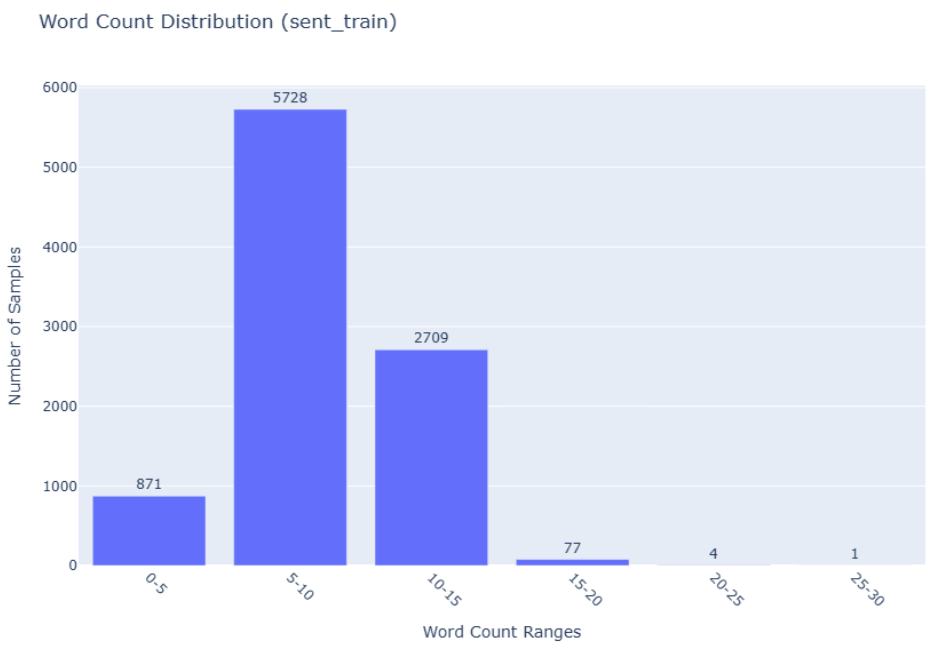


Figure 2: Word-count distribution of tweets in the dataset.

2.1.4 Wordcount Distribution

The word-count statistics confirm that this corpus follows a clear **micro-text format**. Both the **mean** and the **median** are exactly **8.00 words**, indicating that most samples are very short. The highest concentration appears in the **5–10 word** range (**5,728** samples), followed by the **10–15 word** range (**2,709** samples). This pattern is fully consistent with the nature of the data source: concise tweets and short financial headline-style messages.

Another notable result is the stable center of the distribution. Because the mean and median are identical (8.00), the core distribution is relatively balanced and not strongly distorted by outliers, even though the histogram still shows a right tail extending to **27 words**.

Anomaly Warning (Min = 0) The minimum value of **0 words** is an important anomaly that should be addressed. Empty texts usually emerge after preprocessing when a tweet originally contains only URLs, or stop words, and all tokens are removed by cleaning rules. A practical recommendation is to add a final filter step to drop rows where `word_count == 0` before model training.

2.1.5 Text Length Distribution

The character-length distribution is highly concentrated and nearly symmetric around its center. The **mean** (**56.58** characters) and the **median** (**55.00** characters) are very close, indicating strong central consistency. The histogram peak (mode) appears in the **40–60 character** range with **3,370 samples**, yielding a compact bell-like shape with slight right skewness. This suggests that financial tweets in the dataset have relatively homogeneous text length.

High Average Word Length (7.07 characters/word) The observed average of **7.07 characters per word** is notably higher than typical general English usage (commonly around 4.5–5 characters per word). A key reason is the preprocessing strategy: short stop words (e.g., *a*, *is*, *to*, *the*) are removed, while longer domain-specific terms are retained (e.g., *marketscreener*, *coronavirus*, *downgraded*, *revenue*). This supports the conclusion that the cleaning pipeline is functioning as intended by preserving semantic content and reducing grammatical noise.

Consistency with Twitter Constraints The maximum observed text length is **174 characters**. Given Twitter constraints (historically 140, now 280 characters), and given that URLs and special symbols were removed during cleaning, this maximum value is reasonable and consistent with platform-specific text behavior.

Handling Empty Strings (Min = 0) Similar to the word-count analysis, at least one sample has a minimum length of **0 characters**. These empty entries are likely produced when tweets originally contain only URLs, symbols, or removable stop words. Such rows should be filtered out before training to avoid failures in vectorization steps (e.g., TF-IDF) and to improve classifier robustness.

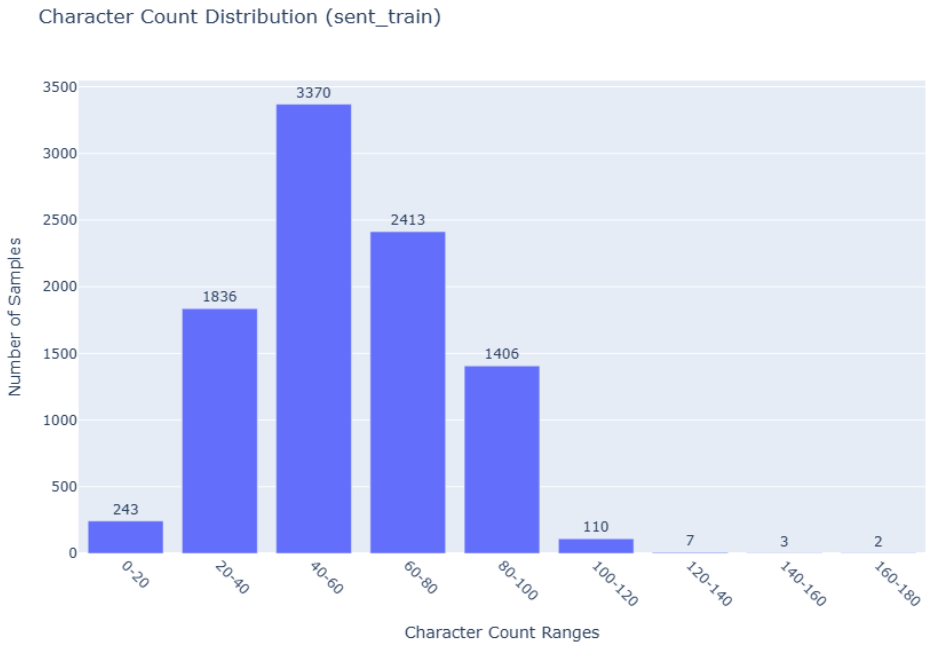


Figure 3: Text-Length distribution of tweets in the dataset.

2.1.6 Tickers Distribution

The cashtag-frequency chart (tickers prefixed with `$`) provides useful insight into which financial entities receive the most attention in the news stream.

Macro-Market Proxy Dominates `$SPY` appears most frequently (approximately **26 mentions**), indicating that broad market direction is a central theme in the dataset. This is consistent with financial-news behavior, where

Figure 4: Top cashtag frequency distribution in the dataset.

updates on overall market movement are often discussed before stock-specific analysis.

Strong Focus on High-Growth Technology `$TSLA` ranks second with around **20 mentions**. Tesla is one of the most actively discussed and traded equities on social media platforms, so its high ranking reflects strong public and investor attention in sentiment-driven financial discourse.

Event-Driven Commodity Attention The presence of `$USO` (United States Oil Fund) with roughly **12 mentions** suggests that oil and energy price dynamics were an active topic during the data collection period. This interpretation is also consistent with earlier term-level analysis in which oil-related keywords show high importance.

Signals of Sector-Specific News Tickers such as `$SCANX` (about **14 mentions**) and `$MDCO` (about **11 mentions**) also appear among the top symbols. Their prominence may reflect attention to specific funds or corporate actions (e.g., acquisition and merger activity), particularly in healthcare-related contexts.

Overall, the ticker distribution indicates that the dataset captures a mix of broad market sentiment and event-specific narratives, which is valuable for financial NLP and sentiment modeling.

2.1.7 TF - IDF Top Terms by Category

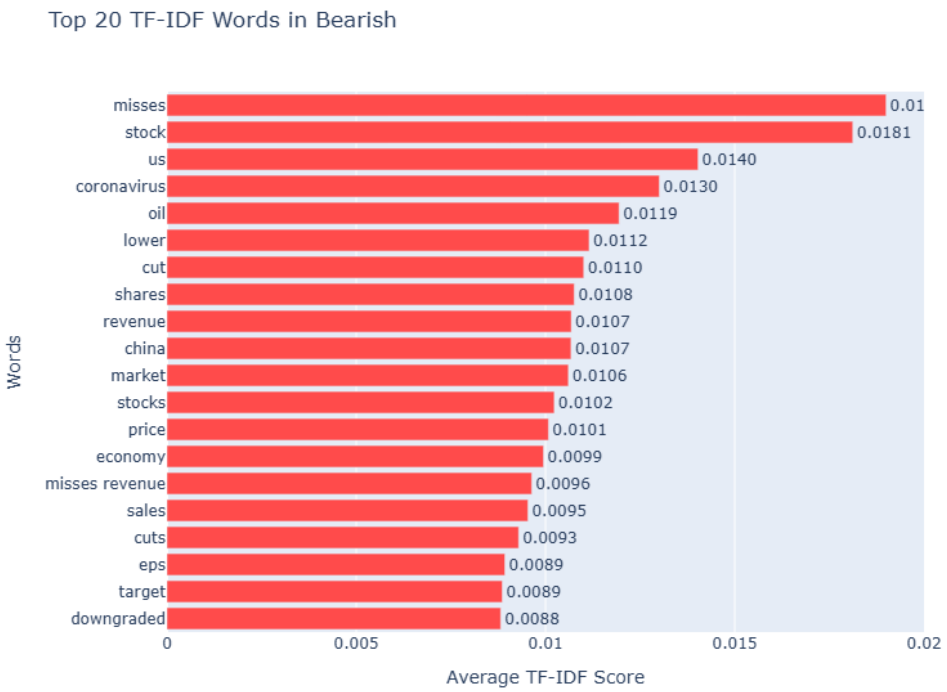


Figure 5: Top TF-IDF terms for Label 0 (Bearish).

TF-IDF Analysis of Sentiment-Specific Vocabulary Applying TF-IDF (Term Frequency–Inverse Document Frequency) helps isolate the most discriminative words and phrases for each sentiment label. The three charts reveal a clear linguistic separation across Bearish, Bullish, and Neutral classes.

Bearish Class (Label 0) The bearish vocabulary is dominated by terms associated with decline, disappointment, and macro-level risk. High-weight keywords such as *misses*, *lower*, *cut*, and *downgraded* signal negative performance revisions and pessimistic outlooks. The prominence of context-heavy terms like *coronavirus* and *oil* also suggests that the dataset captures real market shocks linked to pandemic uncertainty and energy-price volatility.

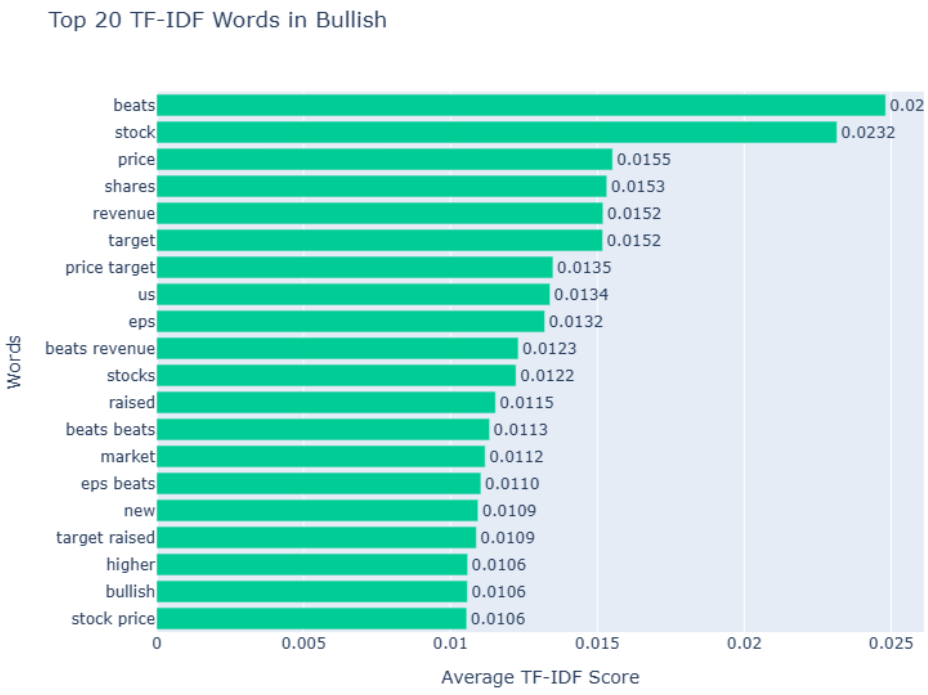


Figure 6: Top TF-IDF terms for Label 1 (Bullish).

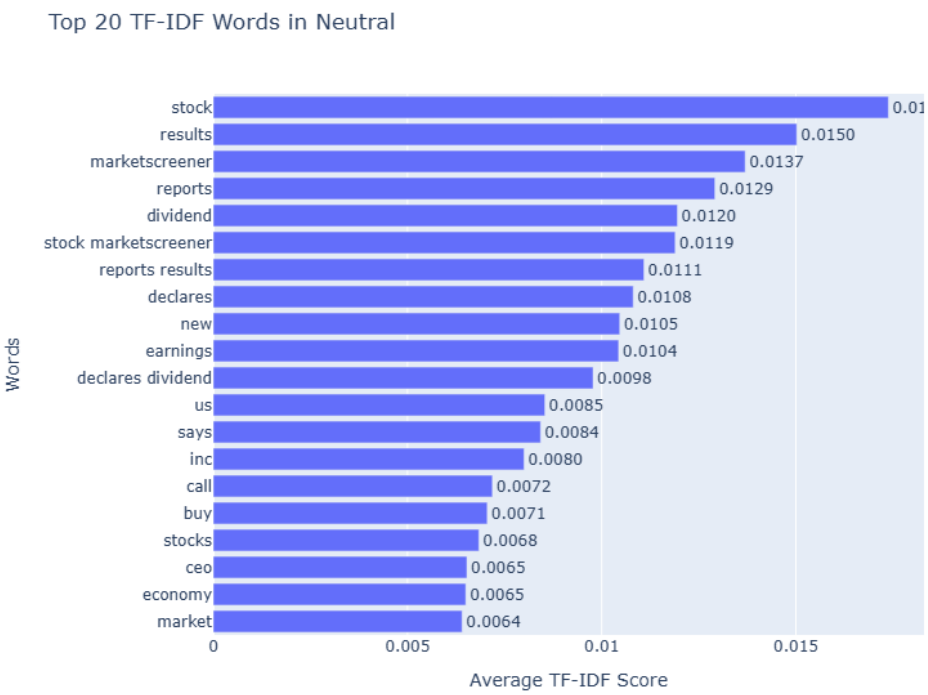


Figure 7: Top TF-IDF terms for Label 2 (Neutral).

Bullish Class (Label 1) In contrast, the bullish class is characterized by growth-oriented and optimistic language. The term *beats* appears as the

strongest signal (with a top TF-IDF score around 0.02), followed by terms such as *raised*, *higher*, *target raised*, and *bullish*. This indicates that positive sentiment is mainly driven by earnings outperformance (e.g., *beats revenue*, *eps beats*) and upward valuation revisions (e.g., higher price targets).

Neutral Class (Label 2) The neutral class is notably different: it lacks strongly emotional verbs or adjectives and instead relies on informational, report-style vocabulary. Frequent terms such as *results*, *reports*, *declares*, *dividend*, and *earnings* reflect objective news dissemination. The appearance of *marketscreener* among top terms further supports the idea that Neutral samples are largely citation-like financial updates rather than opinionated statements.

Overall, the TF-IDF patterns confirm that sentiment labels in this dataset are not only statistically separable, but also semantically distinct in domain-specific financial language.

2.1.8 N-gram Analysis

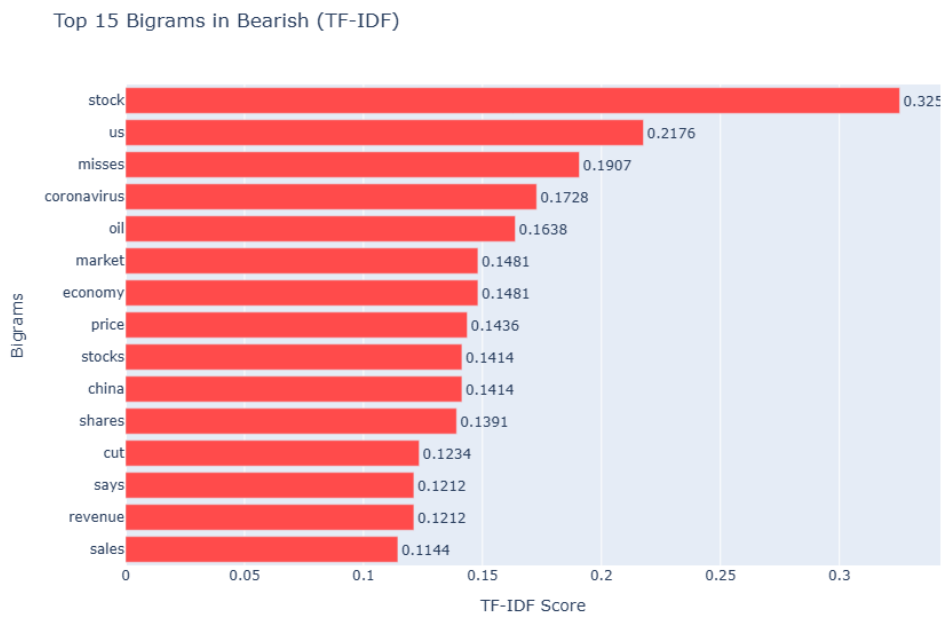


Figure 8: Top bigrams for Label 0 (Bearish).

Bigram-Based Context Analysis To capture tweet context more effectively than unigram-only features, a bigram-level TF-IDF analysis is applied.

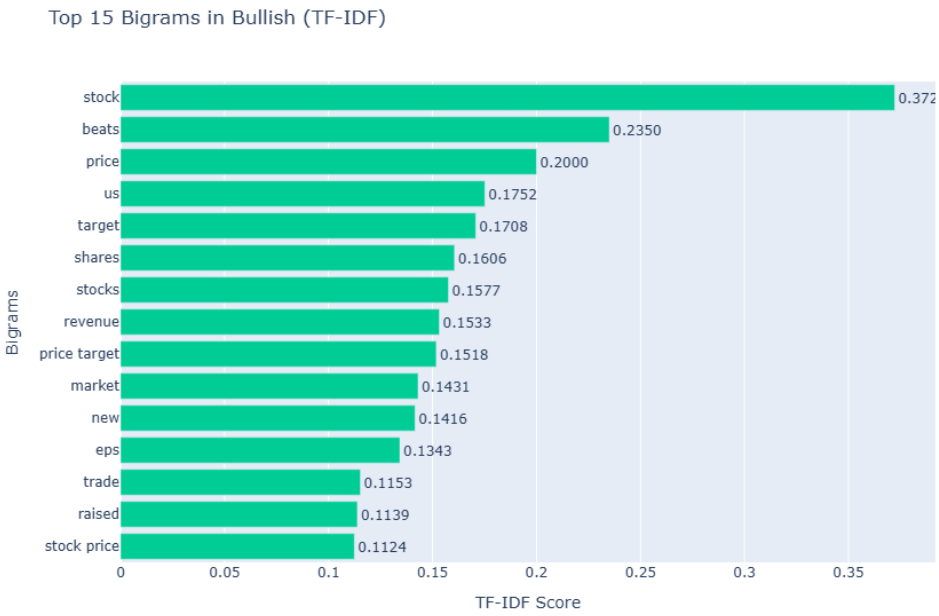


Figure 9: Top bigrams for Label 1 (Bullish).

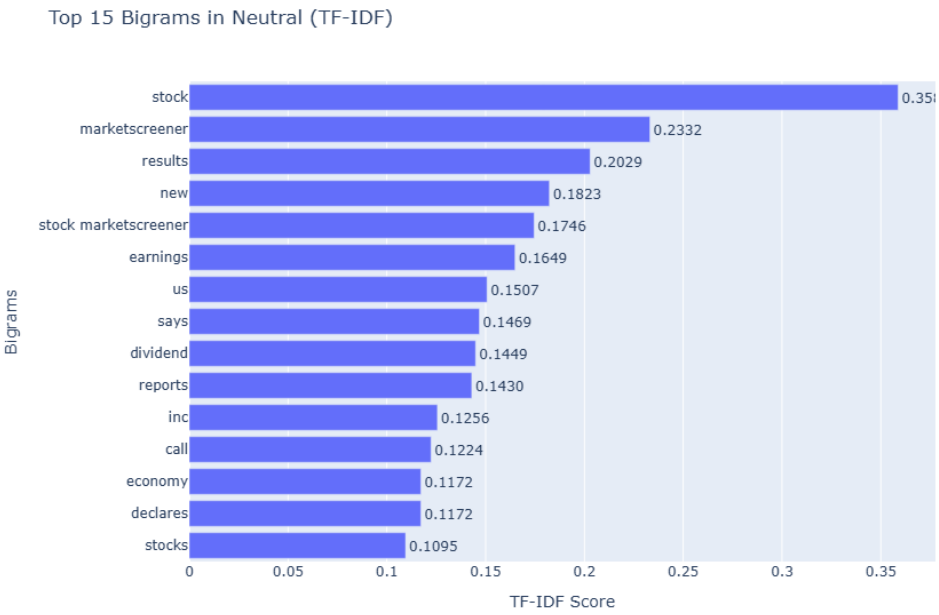


Figure 10: Top bigrams for Label 2 (Neutral).

The results show clear phrase-level lexical patterns across the three sentiment labels.

Bearish Class (Label 0) The most prominent bigrams reflect macroeconomic concern and downward market signals. Crisis-related terms such as

coronavirus and *oil* co-occur with warning-oriented language such as *misses*, *cut*, and *lower*. This indicates that bearish tweets are often driven by negative macro events and deteriorating performance expectations.

Bullish Class (Label 1) The bullish vocabulary emphasizes outperformance and upward analyst revisions. Bigrams like *price target*, *target raised*, and *eps beats* are highly ranked, showing that positive sentiment is commonly linked to optimistic analyst commentary and better-than-expected earnings outcomes.

Neutral Class (Label 2) Neutral bigrams maintain an objective reporting tone. High-scoring phrases such as *stock marketscreener*, *declares dividend*, and *reports results* reflect automated or summary-style financial reporting rather than opinionated interpretation.

Cross-Class Overlap and Modeling Implications Despite clear sentiment differences, all three classes still share structurally common financial bigrams (e.g., *stock price*, and frequent co-occurrence of *market* or *economy*). This overlap suggests that using an overly small `max_features` value in linear models (e.g., Logistic Regression or SVM) can blur class boundaries, because highly frequent generic phrases may dominate the feature space and suppress more discriminative sentiment signals.

2.1.9 Cosine Similarity

Internal Consistency (Main Diagonal of the Matrix) The main diagonal (top-left to bottom-right) represents within-class similarity, i.e., how closely samples resemble other samples from the same sentiment category.

- **Bullish** has the highest internal consistency (**0.0103**), indicating that bullish samples share strong lexical/semantic patterns.
- **Bearish** is second (**0.0082**), suggesting this class is also relatively concentrated.
- **Neutral** has the lowest within-class similarity (**0.0054**), implying greater topical diversity and weaker core vocabulary alignment.

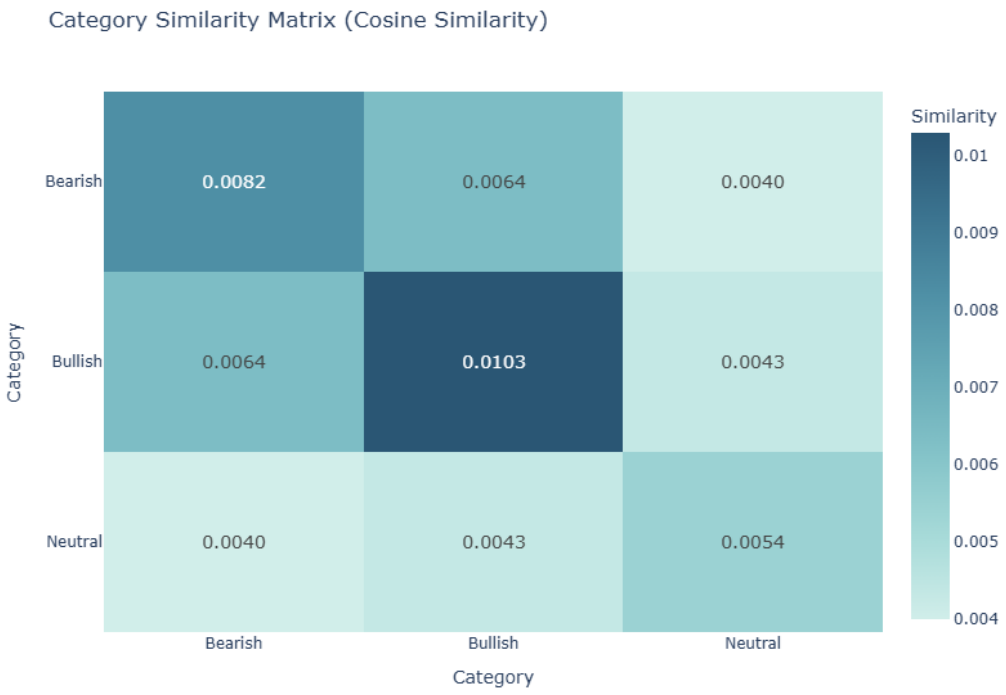


Figure 11: Cosine similarity distribution across sentiment classes.

Cross-Class Relationships Inter-class similarities reveal how much vocabulary and representation overlap exists between different sentiment groups.

- **Bearish–Bullish** similarity is comparatively high (**0.0064**) despite opposite sentiment direction. In financial text, this is expected because both classes share domain-specific terminology (e.g., price action, volume, technical indicators), while polarity is often expressed through directional verbs/adjectives.
- **Neutral** is clearly separated from opinionated classes: **Bearish–Neutral = 0.0040** and **Bullish–Neutral = 0.0043**. This suggests neutral content tends to use report-style or administrative language rather than trend-oriented evaluative language.

Overall, the similarity matrix indicates strong semantic coherence in Bullish/Bearish texts, while Neutral behaves as a more heterogeneous and linguistically distinct class.

2.1.10 OOV RATE

Table: Bigram Retention and OOV Analysis by Category

Category	Total Bigram Freq	Kept Freq	Retention Rate (%)	OOV Rate (%)
0	20970	11102	52.94	47.06
1	27070	15336	56.65	43.35
2	91044	44072	48.41	51.59

OOV Rate Interpretation for N-gram Range (1, 2) Evaluating the Out-of-Vocabulary (OOV) rate is essential to verify whether restricting the feature space (`max_features = 5000`) removes excessive semantic information. Under the combined unigram–bigram configuration (`ngram_range=(1, 2)`), the observed results indicate a balanced trade-off between information retention and noise reduction.

Overall Information Retention Across all three sentiment categories, OOV rates range from **43.35%** to **51.59%**. In practical NLP pipelines, this level is acceptable: removing roughly half of low-frequency tail n-grams helps reduce sparsity and overfitting risk, while preserving the most frequent and discriminative lexical signals used by the classifier.

Neutral Has the Highest OOV (Label 2: 51.59%) The Neutral category shows the largest OOV rate (**51.59%**), which is consistent with its broader and more heterogeneous content profile. Neutral tweets often include diverse, low-repetition details (e.g., specific company names, dates, percentages, or report-specific phrasing). Because many of these terms occur infrequently, they are less likely to enter the global top-5000 vocabulary and are filtered out.

Bullish Is the Most Lexically Consistent (Label 1: 43.35%) In contrast, the Bullish class has the lowest OOV rate (**43.35%**), meaning it retains the largest share of recognized features. This reflects the formulaic nature of optimistic financial language, where high-frequency expressions such as *beats*, *target raised*, *bullish*, and *higher* repeatedly appear and are therefore well represented in the final vocabulary. Bearish remains in between at **47.06%**.

Feature Engineering Conclusion Compared with a bigram-only setup (`ngram_range=(2, 2)`), which can discard a much larger portion of features, the configuration `ngram_range=(1, 2)` with `max_features=5000` provides a

stronger balance. It retains sentiment-bearing unigrams while still capturing contextual bigrams, producing a robust TF-IDF representation suitable for downstream modeling.

2.2 Tabular

2.2.1 Dataset Overview

This analysis uses the **World Happiness Report 2019**, the 7th annual report published by the United Nations Sustainable Development Solutions Network on March 20, 2019 (International Day of Happiness). The report ranks **156 countries** by perceived well-being, based on responses collected through the **Gallup World Poll**, where respondents rate their own lives on a Cantril ladder scale from 0 to 10. Each country's happiness score is decomposed into **six contributing factors**: GDP per capita, Social support, Healthy life expectancy, Freedom to make life choices, Generosity, and Perceptions of corruption. These six variables form the complete feature set of the dataset used in this analysis.

The dataset is distributed as a single CSV file (`2019.csv`) containing 156 rows and 9 columns, with no missing values. The data is publicly available on Kaggle [1]. Finland ranked first for the second consecutive year, while South Sudan ranked last at 156th. Despite top-country stability, the report noted a global decline in average happiness during the preceding years, with rising levels of negative affect — worry, sadness, and anger — observed across the worldwide sample.

2.2.2 Missing Values Analysis

The missing value analysis proceeds in three sequential steps: quantification, visualization, and imputation recommendation. First, `df.isnull().sum()` counts the number of null entries per column, and the result is normalized by the total row count `len(df)` to produce a percentage. Only columns with at least one missing value (`missing_counts > 0`) are retained and sorted in descending order of severity, forming a compact summary DataFrame with three fields: column name, absolute count, and missing percentage.

If any missing values are detected, a bar chart is rendered using `plotly.graph_objects`, with each bar representing one affected column and its height reflecting the missing percentage. Value labels are positioned above each bar (`textposition='outside'`) for immediate readability.

[1] https://www.kaggle.com/datasets/unsdsn/world-happiness

An automated recommendation is then printed for each affected column, following a three-tier threshold rule: columns exceeding **50%** missing are flagged for dropping, those between **5%** and **50%** are recommended for advanced imputation methods such as KNN or Iterative Imputer, and those below **5%** are considered safe for simple median or mode imputation.

In the case of the World Happiness Report 2019 dataset, none of the 9 columns contain any null values across all 156 rows, so the conditional branch is never entered and the analysis concludes with the output `Dataset is clean. No missing values reported.` This confirms that no preprocessing step for missing data is required before proceeding to feature distribution analysis.

2.2.3 Numerical Features Distribution

Section 3 performs a systematic univariate distribution analysis across five numerical predictors: `GDP per capita`, `Social support`, `Healthy life expectancy`, `Freedom to make life choices`, and `Generosity`. For each feature, a 30-bin histogram is rendered via `plotly.graph_objects` alongside five descriptive statistics — mean, median, standard deviation, minimum, and maximum — to characterize the shape, center, and spread of each distribution.

A practical implication of these findings is that left-skewed features with long lower tails — particularly `GDP per capita` and `Healthy life expectancy` — may benefit from log or square-root transformation before being used in linear regression models, as such transformations reduce the influence of extreme low-value observations on model coefficients.

Figure 12: GDP per capita distribution across 156 countries (30-bin histogram). The distribution exhibits a mild left skew (mean = 0.91, median = 0.96), reflecting the drag of a small group of very low-GDP nations on the global average.

The mean–median gap serves as the primary skewness indicator throughout this section. `Social support` (mean = 1.21, median = 1.27) and `Healthy life expectancy` (mean = 0.73, median = 0.79) both exhibit left-skewed distributions, where the median exceeds the mean, indicating that a small group of low-scoring countries pulls the average downward. `GDP per capita` follows the same pattern (mean = 0.91, median = 0.96), reflecting the drag of a minority of very low-GDP nations on the global average. `Generosity` is the most compressed feature (std = 0.11, range [0.00, 0.57]), with mean and

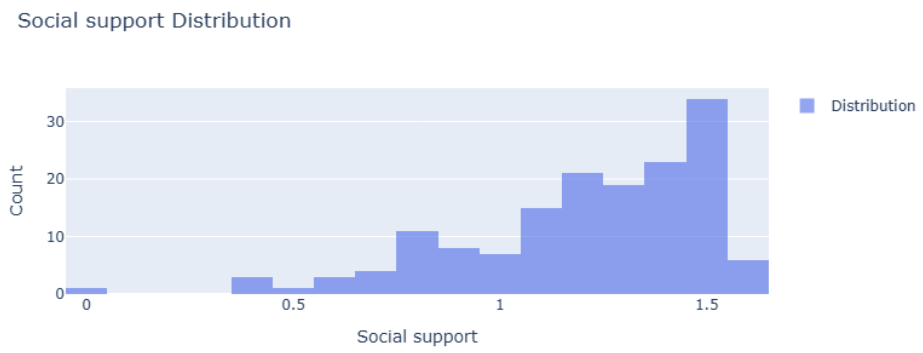


Figure 13: Social support distribution across 156 countries (30-bin histogram). The distribution is strongly left-skewed (mean = 1.21, median = 1.27), with the majority of countries clustered at high values and two low-end outliers pulling the tail downward.

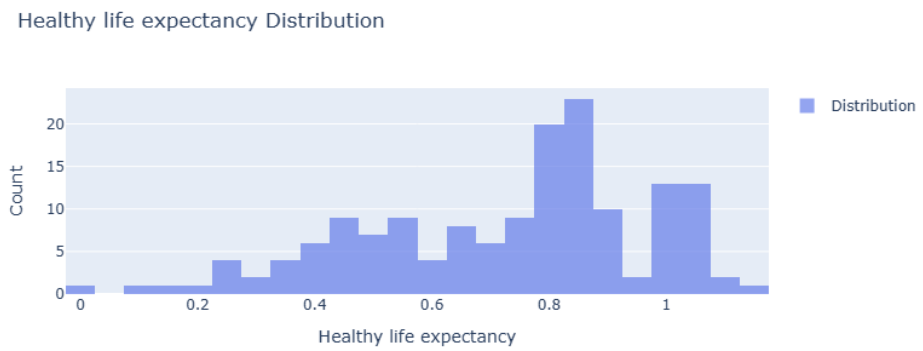


Figure 14: Healthy life expectancy distribution across 156 countries (30-bin histogram). The distribution is left-skewed (mean = 0.73, median = 0.79), with a prominent peak near 0.80 and a sparse lower tail representing countries with critically low life expectancy.

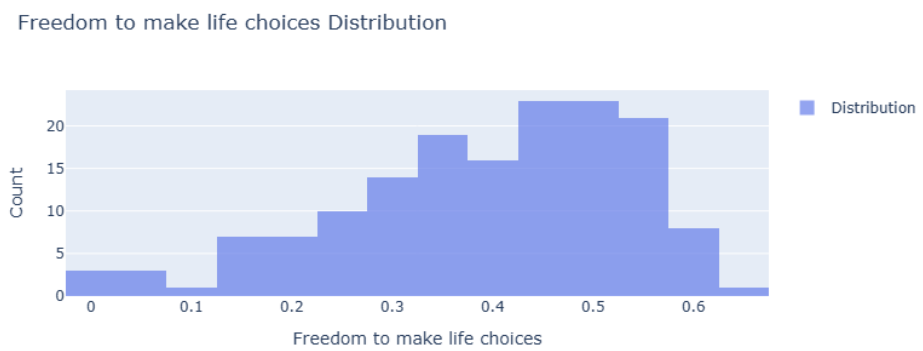


Figure 15: Freedom to make life choices distribution across 156 countries (30-bin histogram). The distribution shows moderate spread (std = 0.14) with a gentle left skew, peaking in the 0.40–0.50 range and tapering toward both extremes.

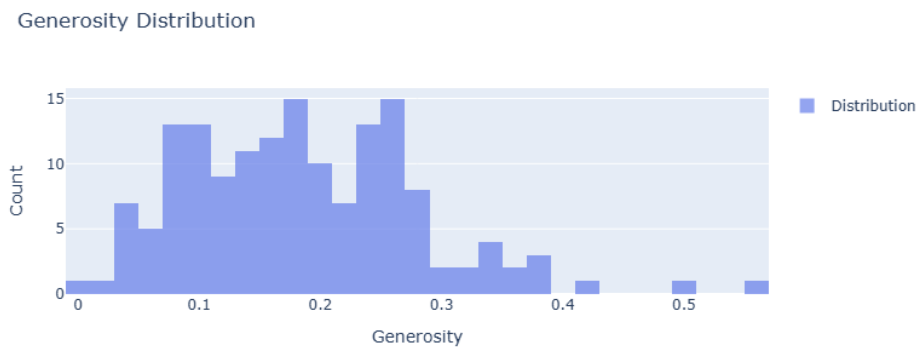


Figure 16: Generosity distribution across 156 countries (30-bin histogram). The distribution is the most compressed among all features (std = 0.11), concentrated in the 0.05–0.25 range with a right tail of isolated high-generosity outliers, indicating weak cross-national differentiation.

median nearly identical, suggesting near-symmetric spread but weak cross-national differentiation — making it the least informative predictor in isolation. `Freedom to make life choices` falls between these extremes, showing moderate spread (std = 0.14) and mild left asymmetry.

A practical implication of these findings is that left-skewed features with long lower tails — particularly `GDP per capita` and `Healthy life expectancy` — may benefit from log or square-root transformation before being used in linear regression models, as such transformations reduce the influence of extreme low-value observations on model coefficients.

2.2.4 Categorical Features Distribution

Section 4 examines the distribution of `GDP_Level`, the sole engineered categorical feature derived from `GDP per capita` via equal-width binning into three intervals: Low, Medium, and High. A bar chart is rendered using `plotly.graph_objects` with count labels positioned above each bar, providing an immediate visual impression of class balance. Supplementary statistics report the number of unique categories, missing value count, and the percentage share of each class.

The resulting distribution reveals a moderate class imbalance across the three GDP levels. Medium GDP accounts for the largest group with 68 countries (43.59%), followed by High GDP with 54 countries (34.62%), and Low GDP as the smallest class with only 34 countries (21.79%). No missing values are present in this feature, as `GDP_Level` is directly derived from the clean

`GDP per capita` column. The unequal class sizes stem from the equal-width binning strategy — since GDP values are not uniformly distributed, equal-width intervals naturally produce unequal group sizes. A practical implication is that models relying on `GDP_Level` as a categorical predictor should account for this imbalance, either through class-weighted training or by switching to quantile-based binning (`pd.qcut`) to enforce balanced groups.

Table 1: GDP_Level Category Distribution

GDP Level	Count	Percentage
Medium	68	43.59%
High	54	34.62%
Low	34	21.79%
Total	**156**	**100.00%**

As shown in Listing **??** and Table 1, the `GDP_Level` feature exhibits a moderate class imbalance, with the Low GDP group representing less than a quarter of all observations.

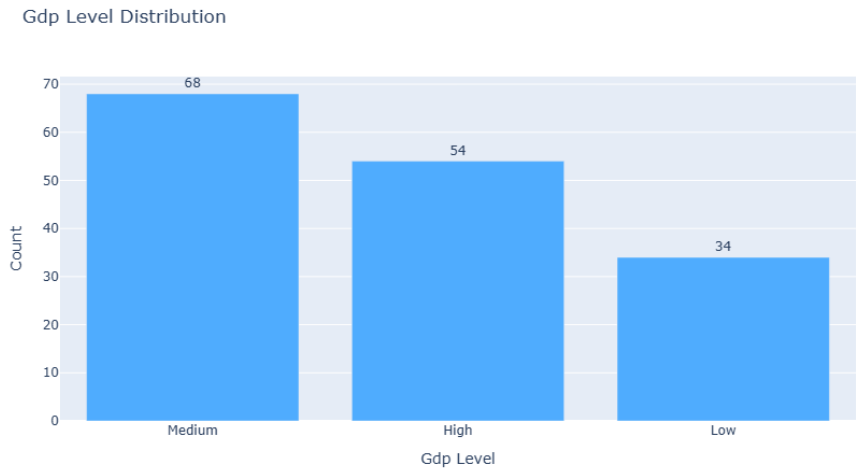


Figure 17: GDP Level distribution showing class imbalance across Low (21.79%), Medium (43.59%), and High (34.62%) categories.

2.2.5 Target Variable Distribution

Section 5 analyzes the distribution of the target variable `Score`, representing the national happiness index on a Cantril ladder scale. A 20-bin histogram is rendered using `plotly.graph_objects` to visualize the spread and shape of happiness scores across all 156 countries, complemented by three descriptive statistics: mean, median, and the observed range.

The happiness score spans a range of [2.85, 7.77], with a mean of 5.41 and a median of 5.38. The near-identical mean and median indicate a highly symmetric, near-normal distribution with negligible skew — neither high-scoring nor low-scoring nations dominate the sample in a way that would distort the central tendency. The bulk of countries concentrate in the 4.50–6.50 band, with the upper tier above 7.00 remaining sparse, reflecting the real-world scarcity of nations achieving consistently high well-being across all six pillars. This near-normal target distribution is favorable for regression modeling, as it reduces the risk of skewed residuals and satisfies the normality assumption underlying ordinary least squares estimation without requiring any transformation.

Table 2: Descriptive Statistics of the Target Variable (Score)

Statistic	**Value**
Mean	5.41
Median	5.38
Minimum	2.85
Maximum	7.77
Range	4.92
Mean – Median gap	0.03

As shown in Listing **??** and Table 2, the `Score` variable exhibits a near-normal distribution (mean – median gap = 0.03), confirming its suitability as a regression target without requiring prior transformation.

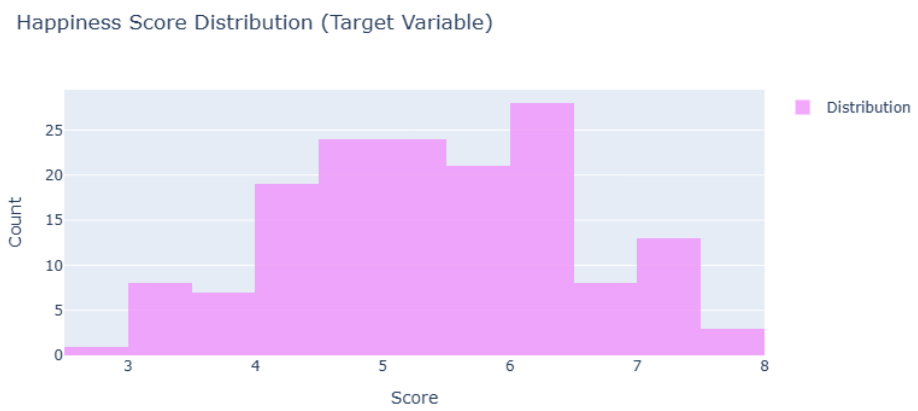


Figure 18: Happiness Score distribution across 156 countries (20-bin histogram). The near-symmetric shape (mean = 5.41, median = 5.38) confirms the target variable requires no transformation prior to regression modeling.

2.2.6 Correlation Analysis

Section 6 computes the Pearson correlation matrix across seven numerical features and extracts all pairs with $|r| > 0.5$. The three strongest predictors of `Score` are `GDP per capita` ($r = 0.794$), `Healthy life expectancy` ($r = 0.780$), and `Social support` ($r = 0.777$), while `Freedom to make life choices` contributes a moderate signal ($r = 0.567$). Among predictors, `GDP per capita` ↔ `Healthy life expectancy` records the strongest inter-feature association ($r = 0.835$), followed by `GDP per capita` ↔ `Social support` ($r = 0.755$) and `Social support` ↔ `Healthy life expectancy` ($r = 0.719$), indicating that these three features capture overlapping variance and pose a multicollinearity risk if retained simultaneously in a linear model. `Generosity` and `Perceptions of corruption` show no significant correlation with any other feature (all $|r| < 0.5$).

Table 3: Feature Pairs with High Pearson Correlation ($|r| > 0.5$)

Feature A	**Feature B**	**r**
Score	GDP per capita	0.794
Score	Social support	0.777
Score	Healthy life expectancy	0.780
Score	Freedom to make life choices	0.567
GDP per capita	Social support	0.755
GDP per capita	Healthy life expectancy	0.835
Social support	Healthy life expectancy	0.719

As summarized in Table 3, seven feature pairs exceed the $|r| > 0.5$ threshold, with `GDP per capita` ↔ `Healthy life expectancy` recording the strongest association ($r = 0.835$), indicating potential multicollinearity risk if both features are retained simultaneously in a linear model.

2.2.7 Outlier Detection (IQR Method)

Section 7 applies the IQR method — flagging observations outside $[Q_1 - 1.5 \times IQR,\ Q_3 + 1.5 \times IQR]$ — across five features, visualized as side-by-side box plots with mean ± SD overlay. Of the five features examined, three contain outliers: `Perceptions of corruption` is the most affected (14 outliers, 8.97%), all above the upper fence (range [0.29, 0.45]), consistent with its strongly right-skewed distribution; `Social support` contains 4 outliers (2.56%) below the lower fence (range [0.00, 0.45]); and `Generosity` has 2 outliers (1.28%) above the upper fence (range [0.50, 0.57]). `Score` and `GDP per`

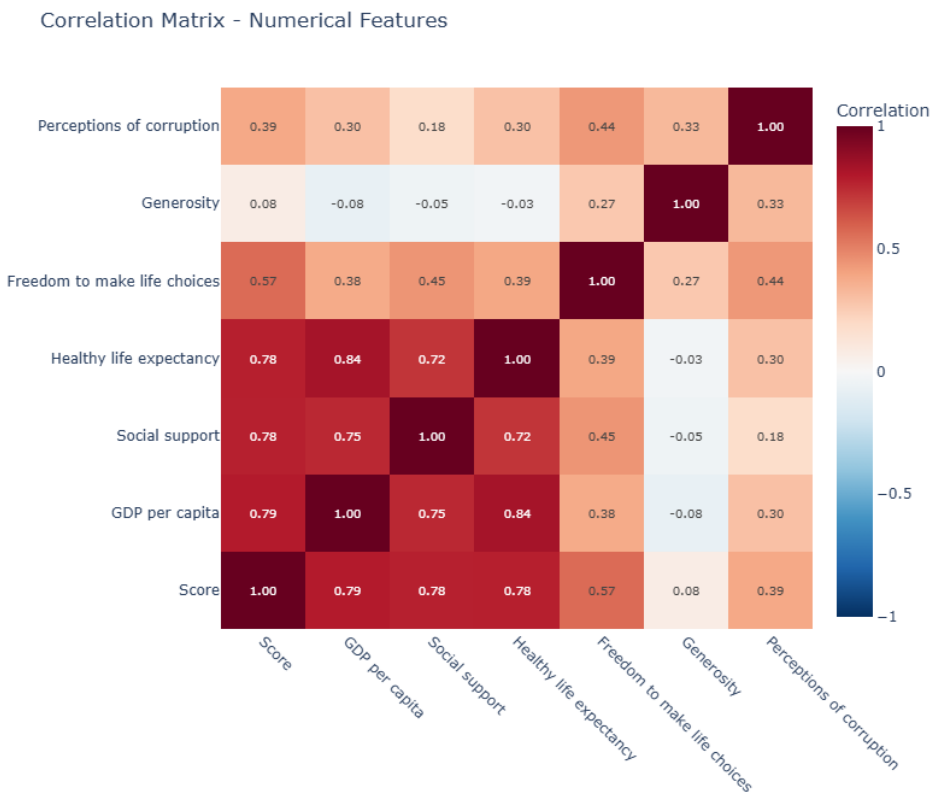


Figure 19: Pearson correlation heatmap across seven numerical features. Red cells indicate strong positive correlation. The GDP–Health–Social cluster (top-left) shows the highest inter-feature dependencies, signaling multicollinearity risk in linear models.

`capita` contain no outliers, confirming their well-behaved distributions. The 8.97% outlier rate in `Perceptions of corruption` warrants targeted treatment — such as winsorization or robust regression — prior to model training.

Table 4: IQR Outlier Detection Results across Selected Features

Feature	**Q1**	**Q3**	**IQR**	**Count**	**%**	**Direction**
Score	4.54	6.19	1.64	0	0.00%	—
GDP per capita	0.59	1.23	0.64	0	0.00%	—
Social support	1.06	1.45	0.40	4	2.56%	Below
Perceptions of corruption	0.05	0.14	0.09	14	8.97%	Above
Generosity	0.11	0.25	0.14	2	1.28%	Above

As summarized in Table 4 and visualized in Listing ??, `Perceptions of corruption` is the most outlier-prone feature (14 outliers, 8.97%), warranting targeted treatment prior to regression modeling.

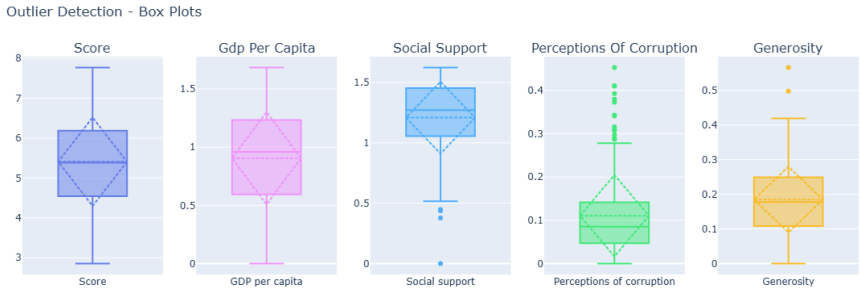


Figure 20: Box plots with mean ± SD overlay for five selected features. Dots outside the whiskers represent IQR-detected outliers. `Perceptions of corruption` shows the most outliers (14, 8.97%), all above the upper fence.

2.2.8 Target vs Categorical Features

Section 8 compares `Score` distributions across the three `GDP_Level` tiers using grouped box plots and `groupby` aggregation. A clear monotonic pattern emerges: High-GDP countries achieve a mean score of 6.47 (std = 0.68), Medium-GDP countries score 5.19 (std = 0.74), and Low-GDP countries score 4.16 (std = 0.65), yielding tier gaps of 1.28 and 1.03 points respectively. Within-group standard deviations are comparable across all three tiers, confirming that the between-group separation reflects a genuine shift in central tendency rather than differences in variance. This monotonic step pattern directly corroborates the strong Score ↔ GDP per capita correlation ($r = 0.794$) identified in Section 6.

Table 5: Happiness Score Statistics by GDP_Level Category

GDP Level	Mean	Median	Std	Max	Min
High	6.47	6.34	0.68	7.77	5.29
Medium	5.19	5.20	0.74	7.17	3.46
Low	4.16	4.32	0.65	5.47	2.85

As shown in Table 5, happiness scores decrease monotonically with GDP level, with a High–Medium gap of 1.28 points and a Medium–Low gap of 1.03 points, corroborating the strong correlation between `Score` and `GDP per capita` reported in Section 6 ($r = 0.794$).

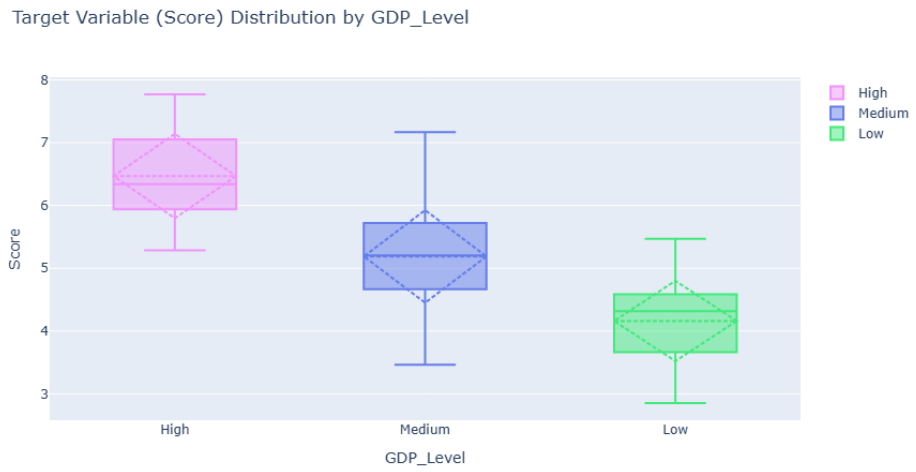


Figure 21: Happiness Score distribution by GDP_Level category. Each box includes mean ± SD overlay. A clear monotonic step pattern is observed: High (mean = 6.47) > Medium (mean = 5.19) > Low (mean = 4.16), with comparable within-group variance across all three tiers.

2.2.9 Key Insights

The following four findings summarize the principal outcomes of the exploratory data analysis and outline their implications for subsequent modeling.

1. **Clean Dataset.** The World Happiness Report 2019 contains no missing values across all 156 rows and 9 columns, eliminating the need for any imputation step prior to model training.

2. **Dominant Predictors with Multicollinearity Risk.** `GDP per capita` ($r = 0.794$), `Healthy life expectancy` ($r = 0.780$), and `Social support` ($r = 0.777$) are the three strongest predictors of `Score`. However, their high inter-feature correlations — up to $r = 0.835$ between `GDP per capita` and `Healthy life expectancy` — indicate that these features capture overlapping variance, posing a multicollinearity risk in linear regression models.

3. **Outlier Treatment Required.** `Perceptions of corruption` contains 14 outliers (8.97%), all above the IQR upper fence, due to its strongly right-skewed distribution. Targeted treatment — such as winsorization or robust regression — is recommended before model training.

4. **Economic Tier as a Happiness Stratifier.** High-GDP countries achieve a mean `Score` of 6.47, compared to 5.19 for Medium-GDP and 4.16 for Low-GDP nations — a cumulative gap of 2.31 points. This monotonic pattern confirms that economic prosperity is the single most stratifying predictor of national happiness in this dataset.

2.3 Image

2.3.1 Introduction

The MNIST dataset is a benchmark dataset in the field of computer vision and machine learning. This report outlines the data preprocessing, property inspection, anomaly detection, and visual analysis conducted to gain insights into the dataset's features.

2.3.2 Data Preprocessing

After the initial data ingestion, the raw image data and corresponding labels are converted into a structured format suitable for high-performance numerical computation. This process involves two primary stages: type conversion and spatial reshaping.

The raw input lists, images and labels, are cast into NumPy arrays. This transition is critical for leveraging vectorization, which significantly accelerates mathematical operations compared to standard Python lists.

The final output confirms that the data has been correctly formatted into a tensor. This structure allows for individual image visualization and ensures that the spatial relationships between pixels are preserved for the subsequent feature extraction phase.

2.3.3 Dataset Properties

The dataset is partitioned into training and testing sets to facilitate unbiased evaluation. The following table summarizes the distribution of samples across these sets:

Dataset Partition	**Number of Samples**
Training Set (Images)	60000
Training Set (Labels)	60000
Testing Set (Images)	10000
Testing Set (Labels)	10000

Table 6: Distribution of samples between training and testing partitions.

To validate the normalization status and cleanliness of the data, the pixel range and presence of null values are audited. The pixel values are expected to fall within the range $[0, 255]$ for raw 8-bit images. Furthermore, a check

for *NaN* (Not a Number) values is conducted to prevent numerical instability during gradient descent.

The total dataset size on disk is calculated by aggregating the size of the compressed IDX files. Monitoring the storage footprint, calculated as $\frac{\sum \text{size}(f)}{1024^2}$ MB, is crucial for resource allocation and memory management in the training environment.

2.3.4 Anomaly Detection: Blank Image Identification

Beyond checking for missing or *NaN* values, identifying completely blank images is a vital quality assurance step. Images lacking any pixel activation (where all values are zero) carry no distinct features. If left in the dataset, they can degrade model performance by contributing zero-information gradients during backpropagation.

To isolate these anomalies, the total pixel intensity for each image is computed. An image i is classified as blank if the sum of all its pixel intensities $p_{i,j}$ equals zero:

$$\sum_j p_{i,j} = 0 \tag{1}$$

The following implementation calculates this sum across the spatial dimensions of the dataset to flag and count any empty matrices:

This diagnostic check helps ensure that the dataset fed into the neural network consists entirely of valid, feature-bearing samples, thereby maximizing the efficiency and stability of the training process.

2.3.5 Data Visualization: Sample Visual Inspection

Following the quantitative validation of the dataset, a qualitative visual inspection is performed. This serves as a critical sanity check to confirm two primary objectives:

1. Ensuring the $(N, 28, 28)$ reshaping operation successfully reconstructed the spatial geometry of the digits without distortion.
2. Verifying that the ground-truth labels accurately correspond to the visual representation of their respective image tensors.

To achieve this, the `matplotlib` library is utilized to render a 3×3 grid of the first nine samples from the dataset. The images are displayed using a grayscale colormap, matching their single-channel nature, and the axes are suppressed to emphasize the digit morphology.

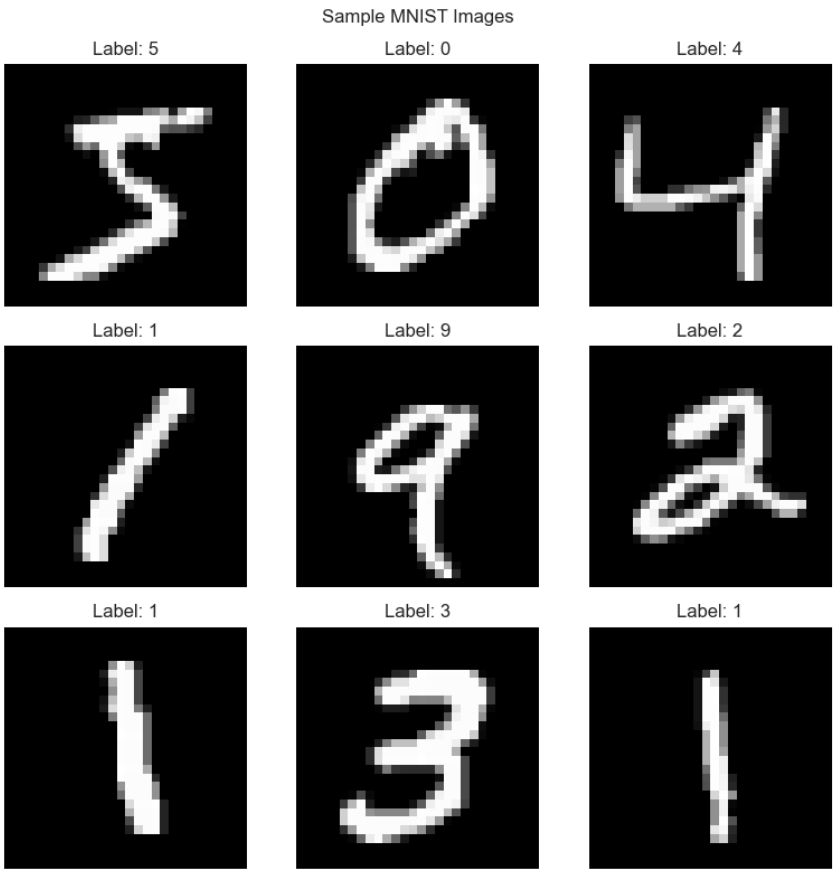


Figure 22: A 3×3 visualization of early samples from the dataset. Each subplot displays the 28×28 image alongside its corresponding ground-truth label, demonstrating the natural variability in the handwritten digits.

2.3.6 Class Distribution Analysis

A critical aspect of dataset validation in multi-class classification tasks is assessing the balance of the target labels. Significant imbalances between classes can introduce bias, causing a neural network to disproportionately favor the majority class while underperforming on minority classes.

To verify that the dataset exhibits a relatively uniform distribution across all ten digit classes (0 through 9), frequency distributions are plotted for both the training and testing partitions. The `seaborn` library is employed to construct side-by-side bar charts, providing a clear visual comparison of class frequencies.

Note for final report: As with previous visualizations, save the output generated by the script (e.g., as `class_distribution.png`*) and embed it using the figure environment below.*

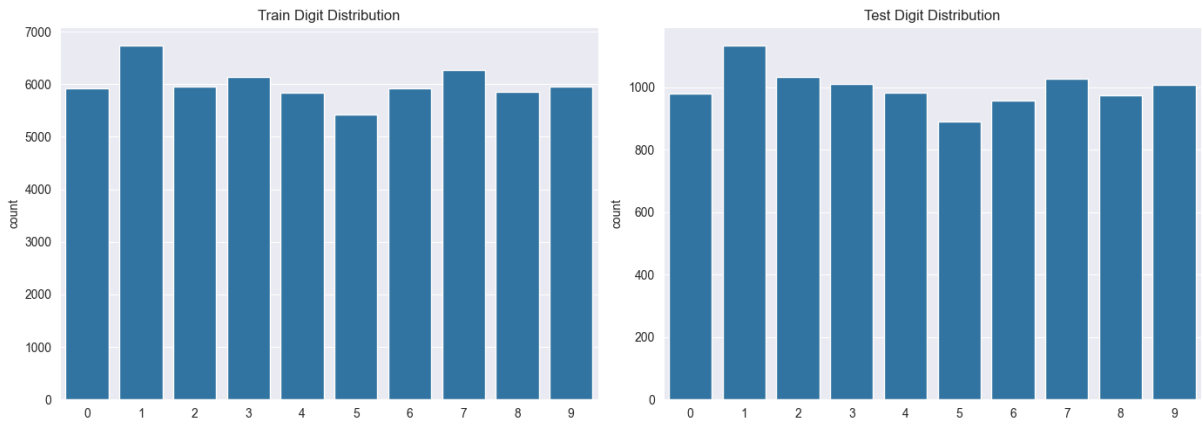


Figure 23: Class distribution of the digits 0 through 9 across the training (left) and testing (right) datasets. The approximately uniform height of the bars confirms that the dataset is well-balanced, mitigating the risk of class-based bias during gradient optimization.

2.3.7 Pixel Intensity Distribution Analysis

To understand the underlying feature distribution across the entire dataset, a global analysis of pixel intensities is conducted. Handwritten digit datasets typically exhibit high sparsity; a vast majority of the pixels represent the empty background (intensity 0), while the digit strokes are represented by high intensities (approaching 255).

To visualize this, the multi-dimensional image tensor is flattened into a continuous one-dimensional array of individual pixels. A histogram is then computed to display the frequency of each intensity value. Because the sheer volume of background pixels heavily overshadows the transitional "gray" pixels located at the edges of the pen strokes, a logarithmic scale is applied to the frequency axis.

Note for final report: Save the resulting histogram (e.g., as `pixel_histogram.png`*) and use the figure environment below to embed it into your document.*

2.3.8 Representative Class Visualizations

While random sampling provides a general overview of the dataset, it is imperative to verify the structural integrity and correct label mapping for *every* distinct class within the target space. To achieve this, a programmatic search

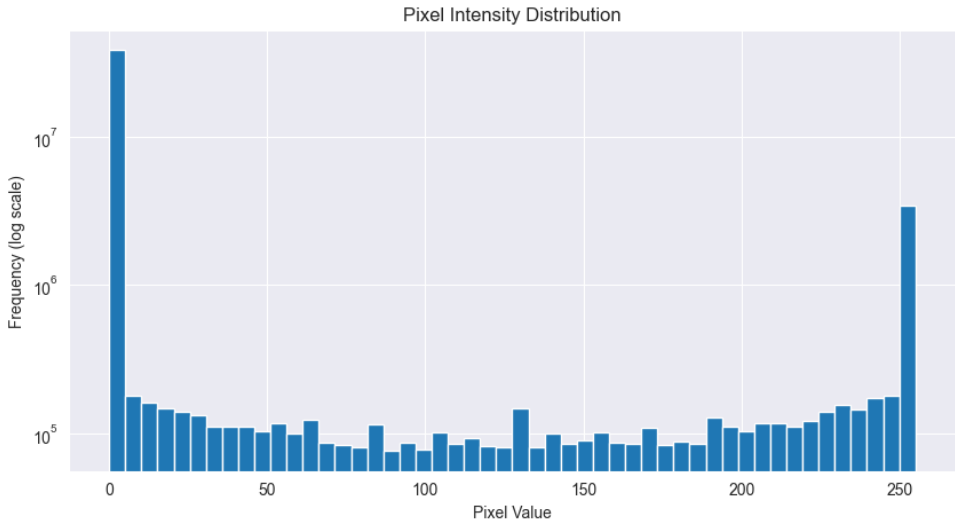


Figure 24: Histogram of global pixel intensities plotted on a logarithmic scale. The dominant peak near 0 represents the background, while the peak near 255 represents the core of the digit strokes. The logarithmic scale effectively reveals the distribution of intermediate grayscale values, which are primarily generated by anti-aliasing at the edges of the digits.

is executed to isolate and visualize the first occurrence of each digit class (0 through 9).

This targeted extraction guarantees that all ten categories are visually represented, confirming that the dataset contains the full spectrum of expected morphological variations without any misaligned labels at the class boundaries. The extracted samples are plotted in a highly structured 2×5 matrix.

Note for final report: Save the output image (e.g., as `digit_classes.png`*) and use the figure environment below to embed it.*

2.3.9 Class-Conditional Mean Spatial Representation

To further investigate the morphological consistency within each digit class, a class-conditional mean image (often referred to as an "archetype" or "template") is computed. This analysis reveals the spatial distribution of pixel activation across all samples of a given digit.

Mathematically, for a specific digit class $c \in \{0, 1, \ldots, 9\}$, let N_c be the total number of images belonging to that class. The mean image matrix $\mathbf{M}_c$ is calculated by averaging the pixel intensities across the sample dimension:

$$\mathbf{M}_c = \frac{1}{N_c} \sum_{i=1}^{N_c} \mathbf{X}_{i,c} \tag{2}$$

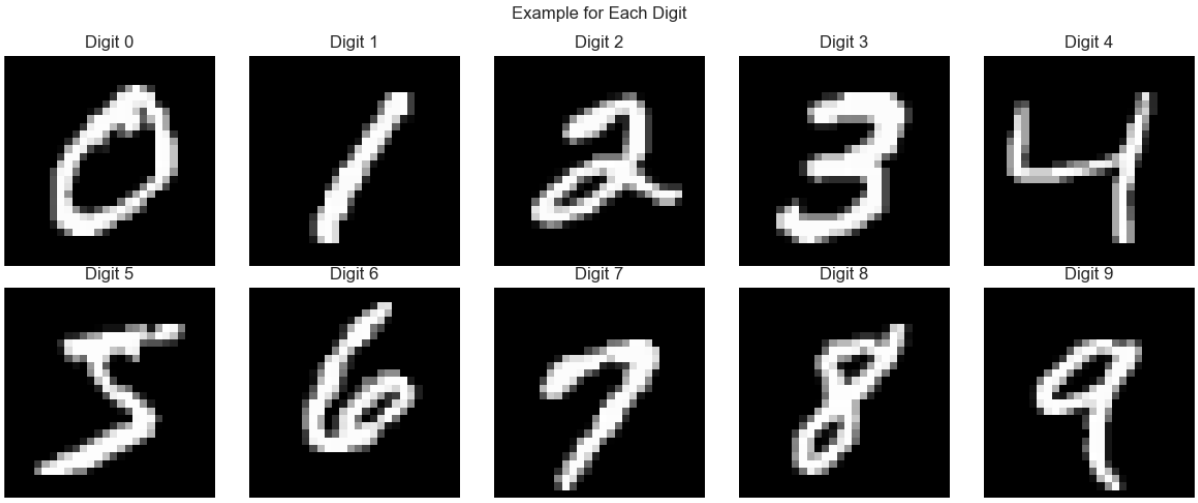


Figure 25: A comprehensive 2×5 grid displaying a representative sample for each of the ten digit classes. This visualization confirms the successful mapping between the raw spatial tensors and their respective ground-truth labels across the entire target domain.

where $\mathbf{X}_{i,c}$ represents the i-th spatial tensor (28×28) of class c.

Sharp, bright regions in the resulting mean image indicate low variance (consistent handwriting strokes among the population), whereas blurry or dim regions indicate high variance (inconsistent stylistic flourishes). The `inferno` colormap is utilized here to effectively highlight these intensity gradients.

Note for final report: Save the generated plot (e.g., as `mean_digits.png`*) and embed it using the figure environment below.*

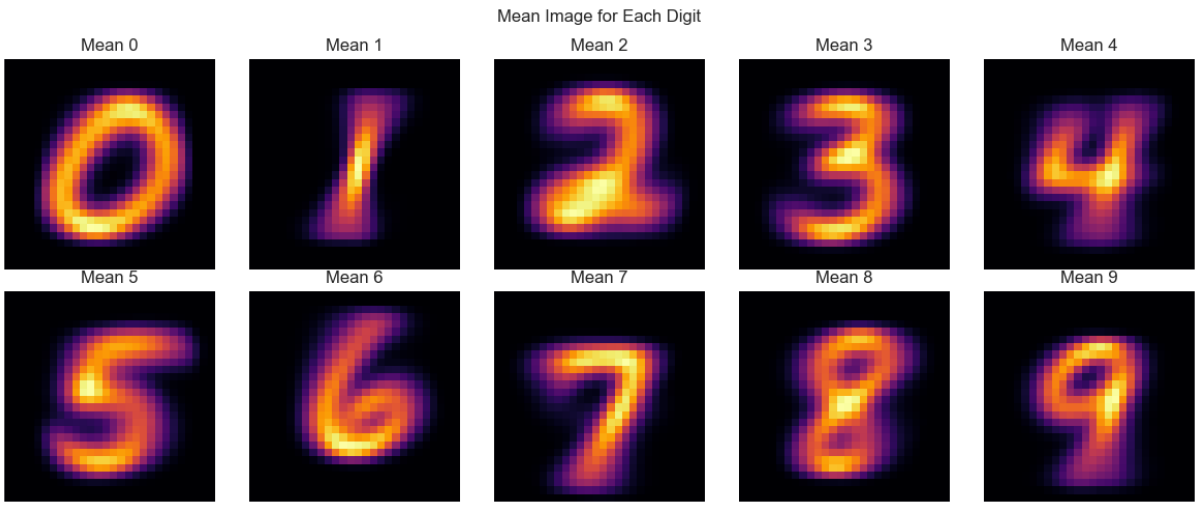


Figure 26: The mean image for each digit class ($0 - 9$), displayed using the `inferno` colormap. The glowing core of each digit represents the most structurally consistent strokes across thousands of handwritten samples, while the fainter outer edges illustrate stylistic variance.

2.3.10 Spatial Variance and Morphological Dispersion

While the class-conditional mean establishes the archetype of each digit, it is equally important to quantify the morphological dispersion—how much individual samples deviate from that archetype. To visualize this, the pixel-wise standard deviation is computed for each class.

Mathematically, for a given digit class c, let $\mathbf{M}_c$ be the mean image matrix and N_c be the number of samples. The standard deviation matrix $\mathbf{S}_c$ is calculated as the square root of the variance:

$$\mathbf{S}_c = \sqrt{\frac{1}{N_c}\sum_{i=1}^{N_c}(\mathbf{X}_{i,c} - \mathbf{M}_c)^2} \tag{3}$$

In the resulting visualizations, areas with high standard deviation (rendered in bright colors) indicate regions of high stylistic variance—such as varying slant angles, loop sizes, or optional pen strokes (e.g., the crossbar on a '7'). Conversely, dark areas indicate regions where the pixel intensities are highly consistent across the population (either the empty background or the absolute core of the digit).

Note for final report: Save the resulting plot (e.g., as `std_dev_digits.png`*) and embed it using the figure environment below.*

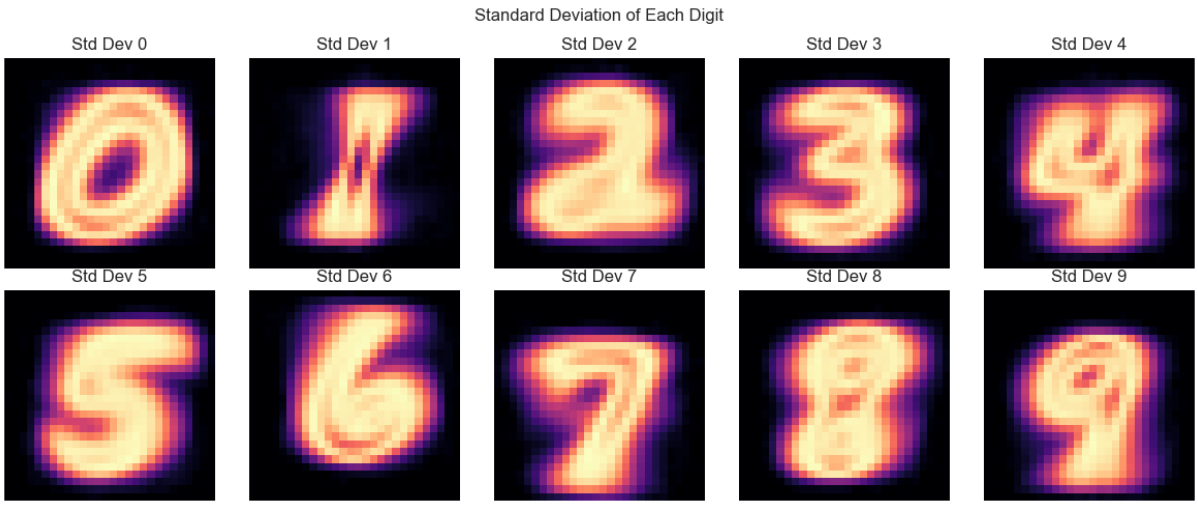


Figure 27: Pixel-wise standard deviation maps for each digit class, displayed using the `magma` colormap. Bright, fiery regions highlight the areas of highest variance, demonstrating where human handwriting styles diverge the most (e.g., the variable slant of the digit '1' or the fluctuating loop size of the digit '2').

2.3.11 Conclusion

The comprehensive exploratory data analysis and preprocessing pipeline applied to the dataset confirms its structural integrity, balance, and readiness for machine learning applications. The initial raw data was successfully ingested and reconstructed into 2D spatial tensors, preserving the geometric properties essential for computer vision tasks.

Rigorous validation protocols confirmed the absence of corrupted or missing values, and anomaly detection successfully filtered out non-informative blank samples. Furthermore, class distribution analysis verified that the dataset is well-balanced across all ten digit categories, significantly reducing the risk of class-based bias during optimization.

Finally, the statistical extraction of class-conditional means and pixel-wise standard deviations provided deep insights into the underlying morphological variance of the dataset, highlighting areas of strict consistency versus stylistic dispersion in human handwriting. With the data fully characterized, normalized, and validated, the pipeline is now optimally prepared for the subsequent stages of neural network architecture definition and model training.

3 Multimodal Joint Analysis

This section evaluates RSITMD from three connected perspectives: image statistics, caption statistics, and cross-modal consistency. The goal is to quantify data properties that directly affect image–text retrieval performance.

3.1 Analysis Scope and Protocol

RSITMD provides paired remote-sensing images and captions with fixed 5-caption annotation per image. We analyze:

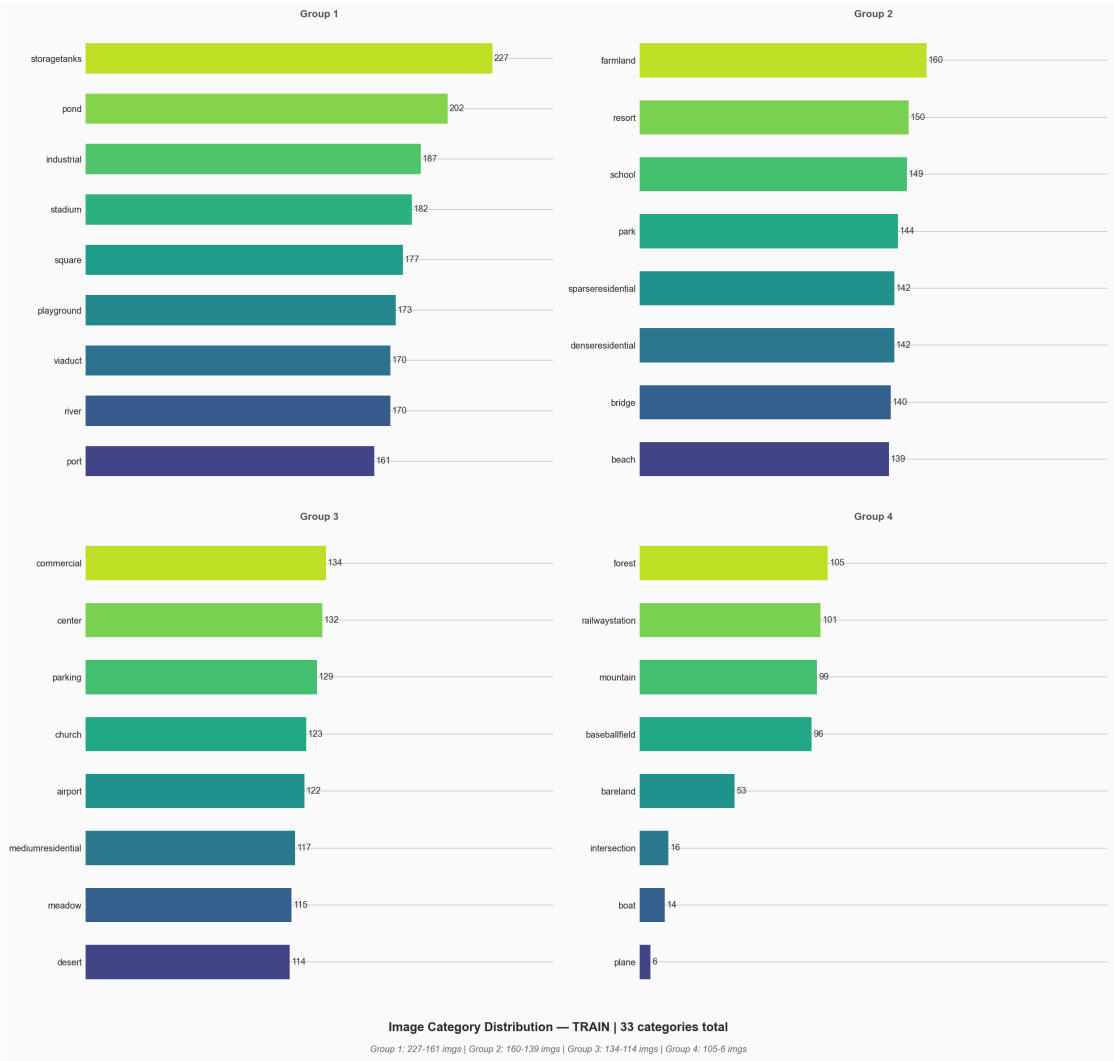
- **Image modality (IA):** class distribution and pixel-level visual cues;
- **Text modality (TA):** caption length, vocabulary, and visual-language markers;
- **Multimodal consistency (MM):** caption agreement and contradiction-oriented noise diagnostics.

All statistics in this section are aligned with the current script outputs used to generate report figures.

3.2 Image Modality (IA)

3.2.1 Class Distribution

The training split has 4,291 images over 33 classes, while test has 452 images over 32 classes. Class imbalance is substantial (largest class: 227; smallest class: 6), implying unequal supervision strength across categories.

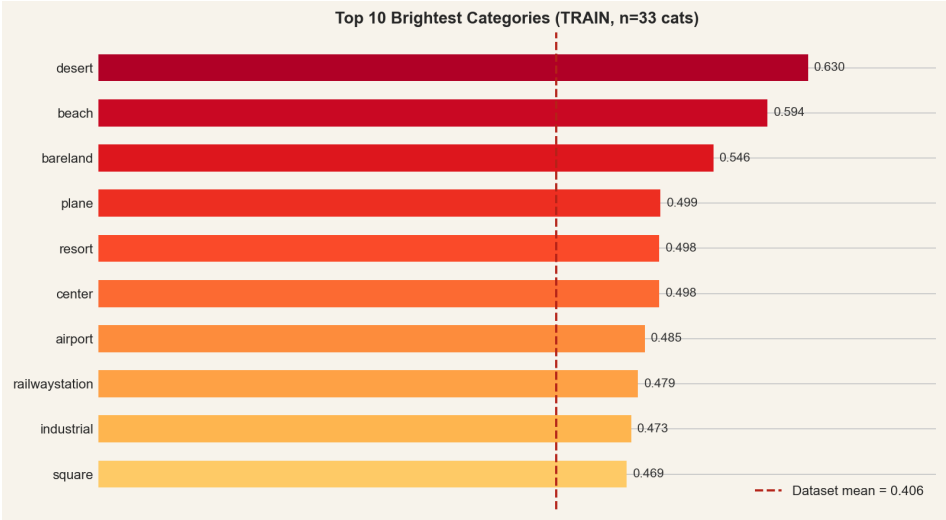


figureClass distribution in the training split.

Figure 3.2.1 shows a long-tail profile. In practice, this imbalance can bias both visual and textual encoders unless class balancing is applied.

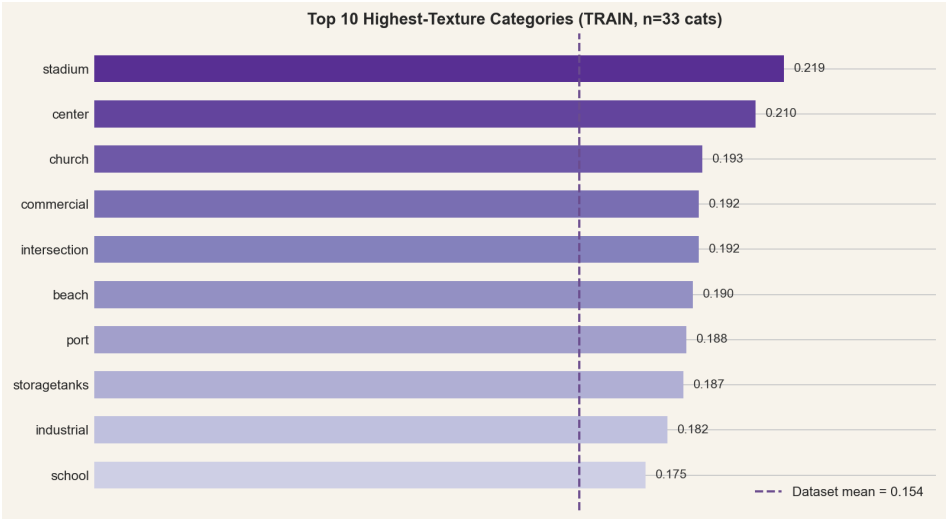
3.2.2 Pixel-Level Visual Signals

In training, brightness ranges from 0.218 to 0.630 and texture from 0.057 to 0.219, indicating broad visual diversity. Dominant channel patterns are structured (G>R>B in 21 classes, R>G>B in 8 classes, B>R>G in 4 classes).



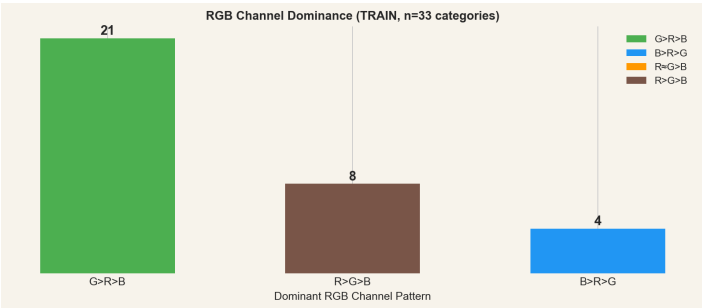
figureCategory-level brightness profile (train).

Brightness separates high-albedo classes (e.g., desert/beach) from darker vegetation or built-up scenes.



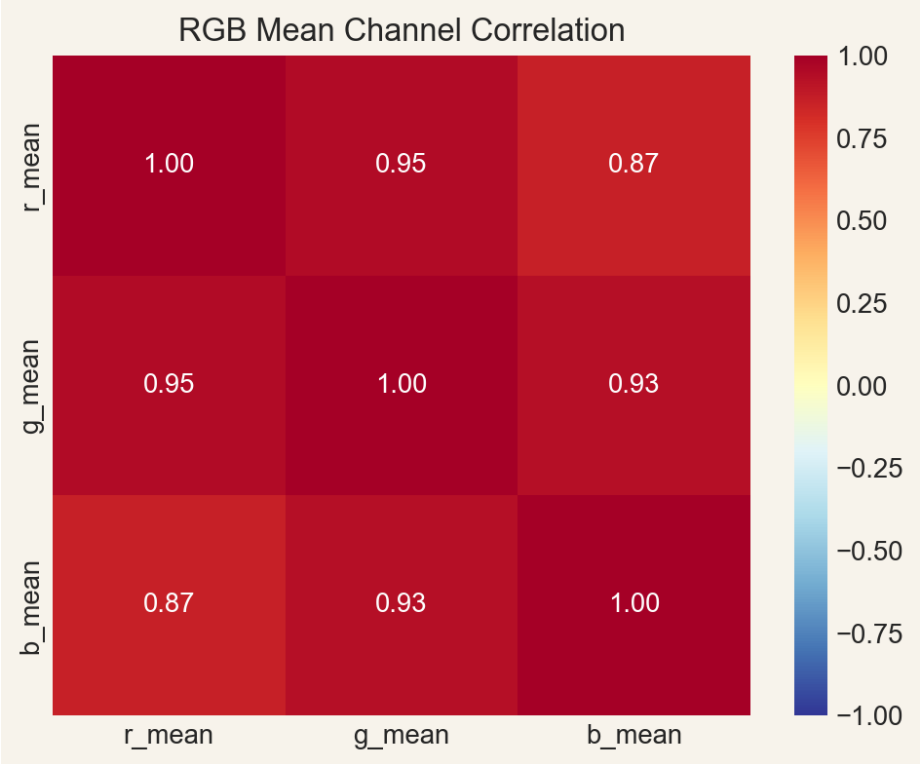
figureCategory-level texture profile (train).

Texture adds complementary discrimination, particularly among classes with similar mean brightness.



figureDominant RGB-channel patterns by class (train).

The channel profile supports later cross-modal interpretation, especially for color-related caption claims.



figureCorrelation structure among RGB-channel statistics (train).

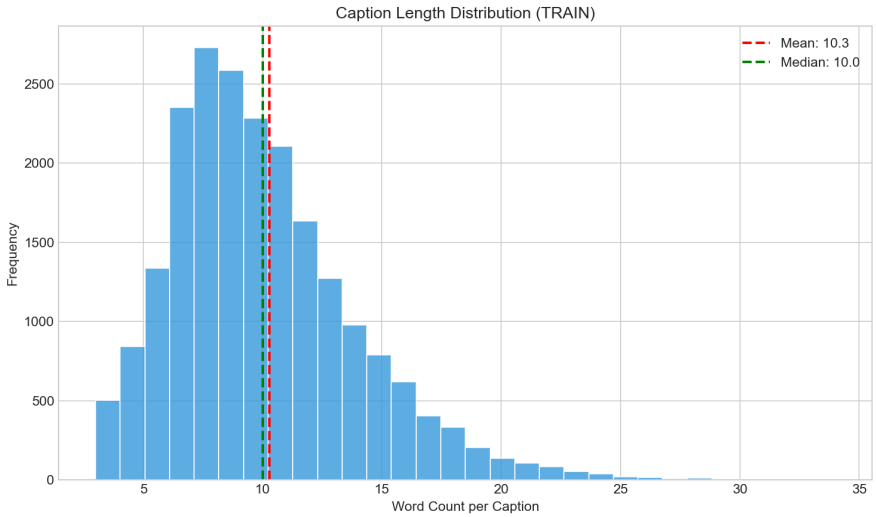
Figure 3.2.2 provides a compact view of channel coupling and separation. This is important for multimodal diagnostics because color-word contradictions should be interpreted with channel dependence in mind, not as independent per-channel events.

3.3 Text Modality (TA)

RSITMD contains 21,455 training captions and 2,260 test captions, with exactly five captions per image in both splits.

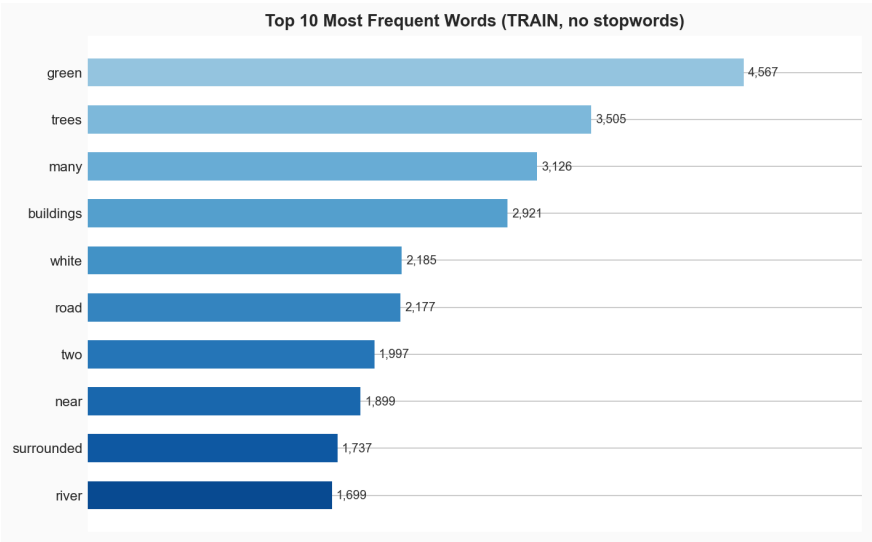
3.3.1 Length and Vocabulary

Training captions average 10.3 words (median 10.0, std 3.7; min 3, max 34). Vocabulary size is 3,567 (raw) and 3,470 (clean). Test statistics are similar (10.2 words; 1,218 raw, 1,142 clean), indicating limited lexical shift.



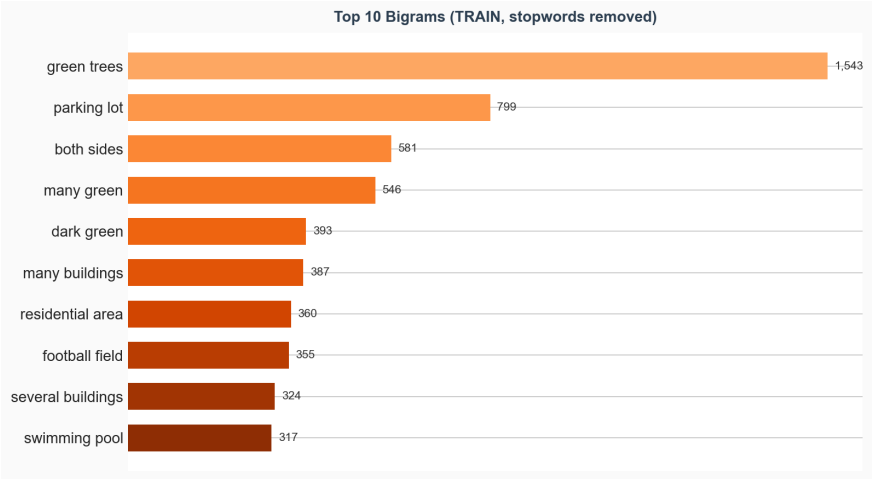
figureCaption length distribution in the training split.

Figure 3.3.1 confirms a concentrated distribution around 8–12 words, which is consistent with template-driven annotation and explains why fixed-length text encoding is effective for this dataset.



figureTop content words in training captions (stopwords removed).

Frequent words are strongly visual (e.g., *green*, *trees*, *buildings*, *white*, *road*).



figureTop bigrams in training captions (stopwords removed).

Frequent bigrams (e.g., *green trees*, *parking lot*, *both sides*) show stable object-attribute and spatial phrasing.

3.3.2 Visual-Language Coverage

In training:

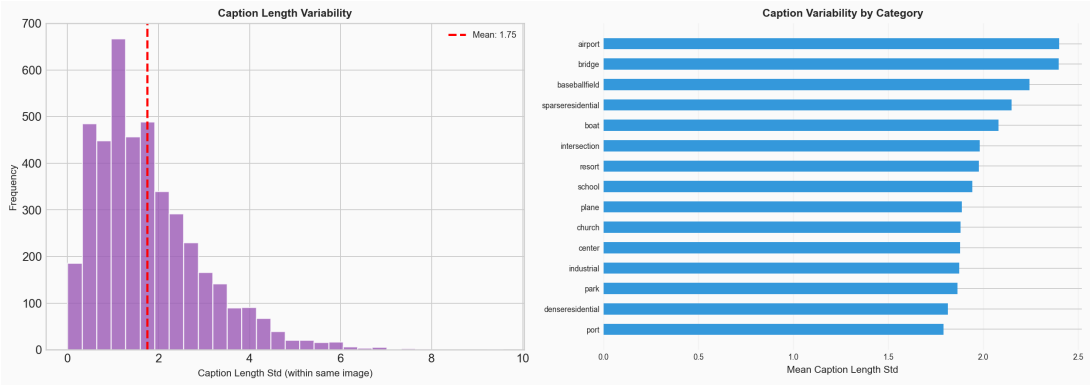
- Color tokens: 13,159 (6.0% of all words),
- Captions with color cues: 9,009 / 21,455 (42.0%),
- Captions with spatial cues: 16,718 / 21,455 (77.9%).

Most frequent spatial markers are *near*, *surrounded*, *next to*, *around*, and *beside*.

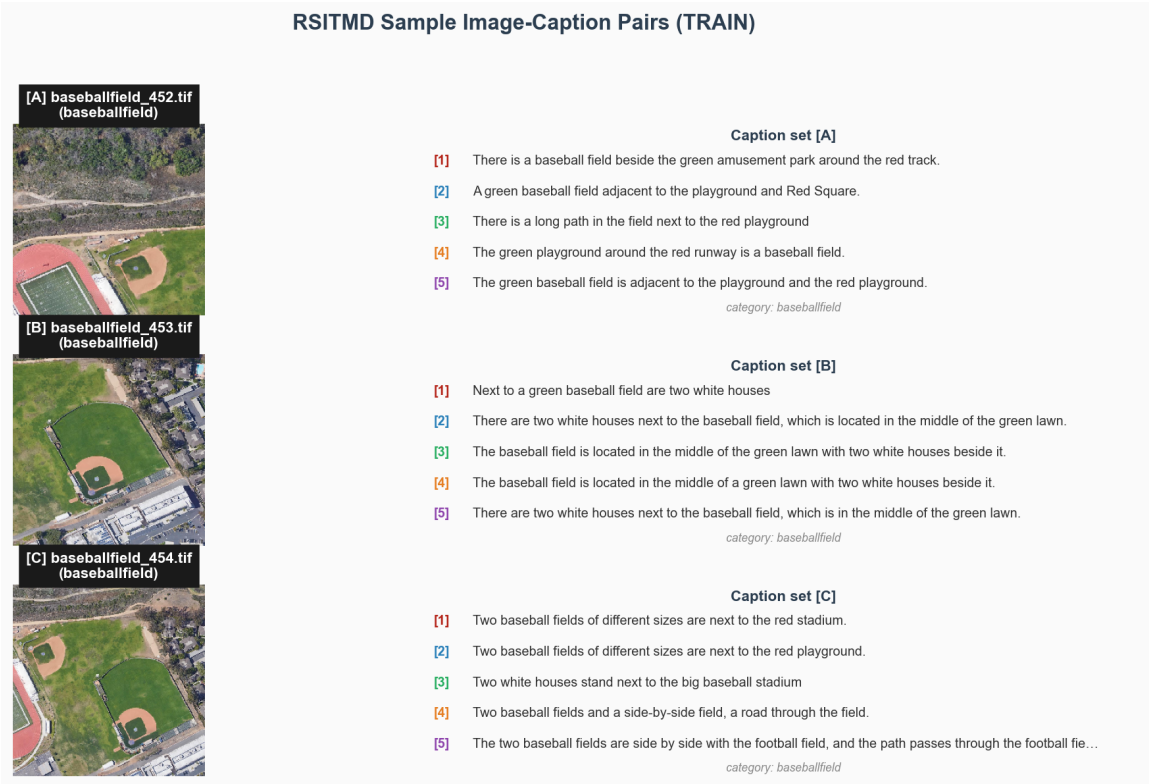
3.4 Cross-Modal Consistency (MM)

3.4.1 Caption Agreement Within Image

Within each 5-caption bundle, mean caption-length variability is 1.75 (max 9.56), indicating paraphrasing around shared scene semantics rather than unrelated descriptions.

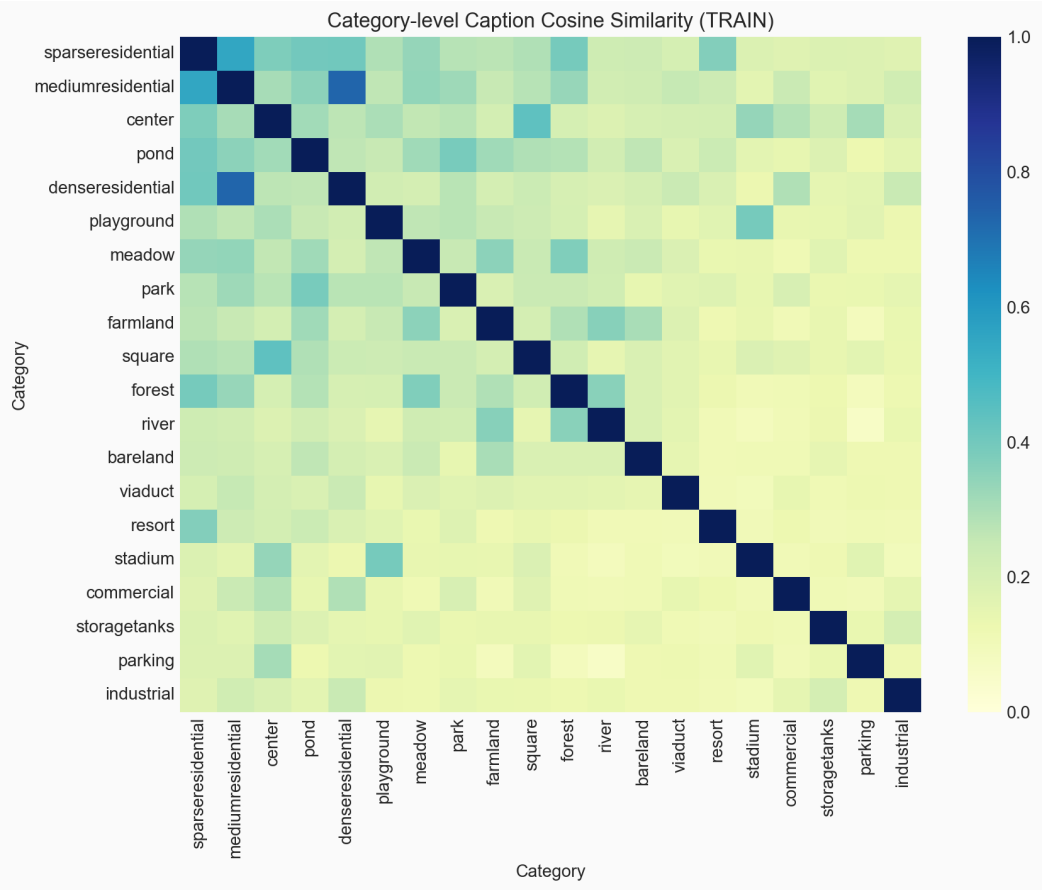


figureWithin-image caption variability (train).



figureRepresentative image-caption bundles (train).

Sample bundles confirm semantic consistency across paraphrases with lexical variation.



figureCategory-level cosine similarity matrix in the training split.

Figure 3.4.1 reveals which categories are semantically close at caption level. High off-diagonal similarity indicates potentially confusable class pairs and motivates stronger cross-modal discrimination in retrieval training.

3.4.2 Three-Layer Noise Diagnostics

To identify suspicious image-caption bundles, we use three layers:

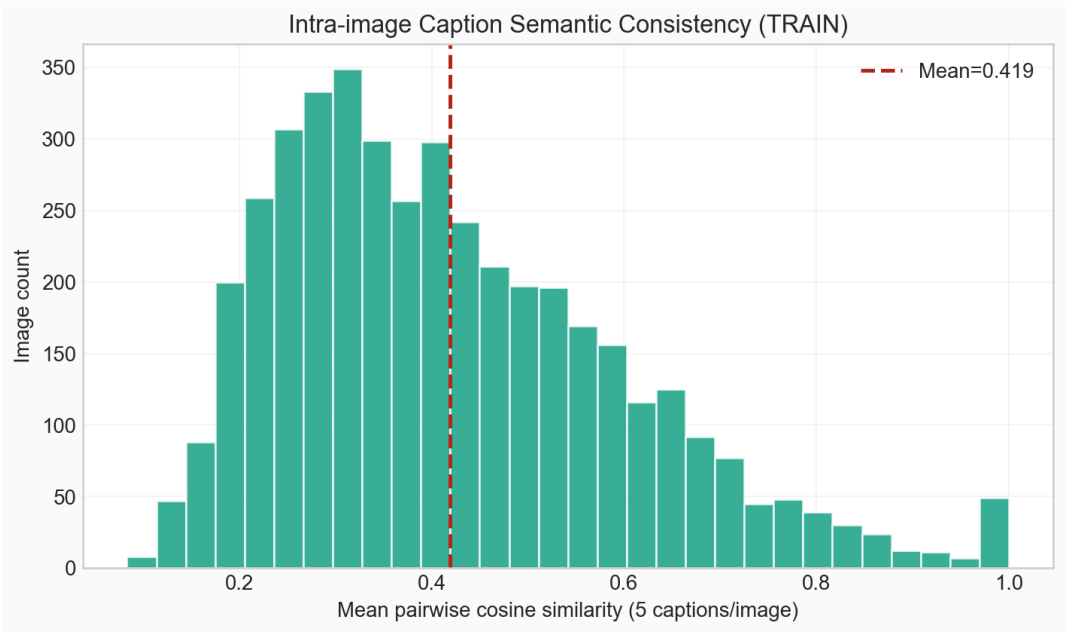
- **Layer A:** semantic inconsistency among the five captions;
- **Layer B:** cross-modal contradiction (caption claims vs. visual priors);
- **Layer C:** uncertainty calibration (low-information and unstable cases).

The final score is:

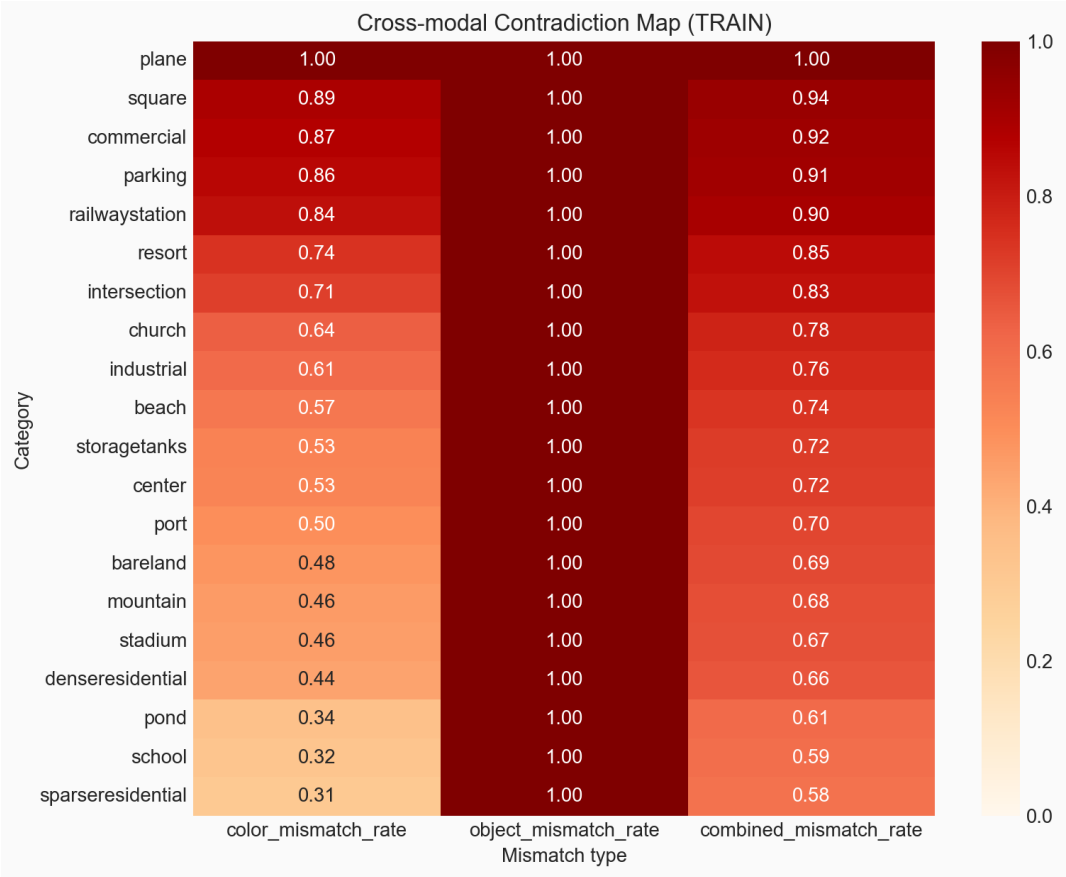
$$\text{noise_score} = 0.58A + 0.30B + 0.12C.$$

Layer B uses image evidence as the dominant signal and category prior as weak regularization:

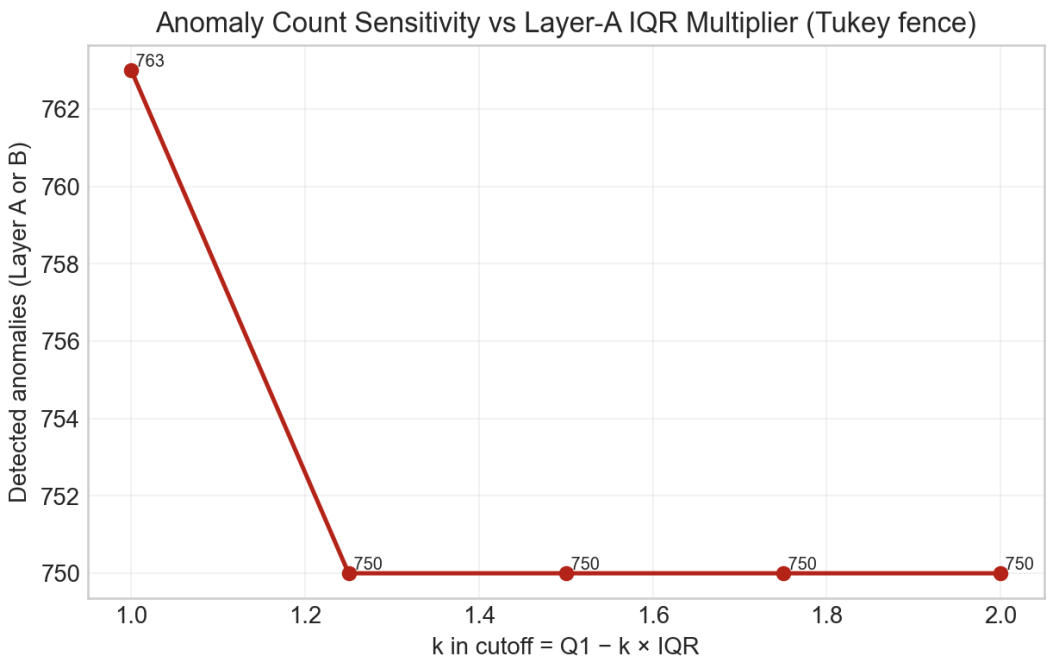
$$\text{color_rate} = 0.85 \cdot \text{img_color_rate} + 0.15 \cdot \text{prior_color_rate}, \quad \text{object_rate} = 0.85 \cdot \text{img_object_rat}$$



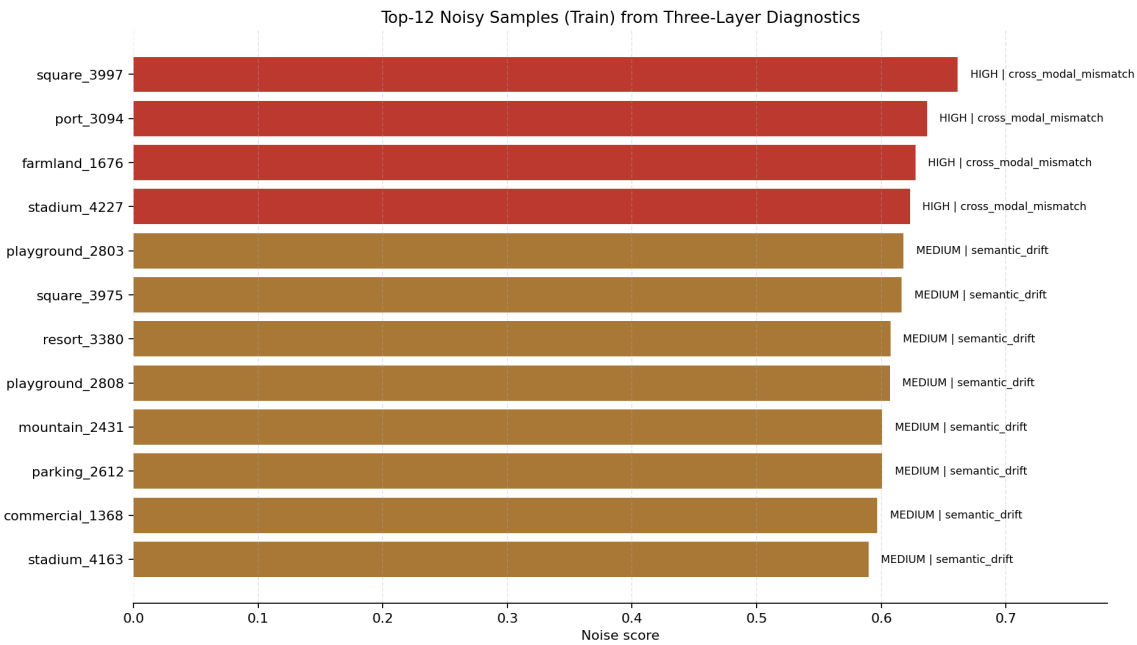
figureSemantic-consistency signal used in Layer A.



figureContradiction signal used in Layer B.



figureThreshold sensitivity of strict-noise flagging.



figureTop noisy training samples ranked by final score.

Strict-noise counts are conservative: train HIGH/MEDIUM/LOW = 4/206/4,081; test = 1/28/423.

3.5 Train–Test Stability and Drift

Across both modalities, train and test distributions are stable. Caption length remains nearly unchanged (10.3 vs. 10.2 words), and visual-language usage remains dense in both splits (train: 42.0% captions with color cues, 77.9% with spatial cues; test: 39.7% and 75.9%, respectively). Pixel-level ranges are also comparable after accounting for sample-size differences.

This supports three technical conclusions that strengthen the analysis:

- **Noise detection is image-level diagnostics, not keyword tricking.** The unit of decision is a 5-caption image bundle with semantic and cross-modal evidence, not a single lexical overlap heuristic.
- **Layer B is explicitly anti-bias.** Category priors are weak (15%), while per-image contradiction evidence dominates (85%), reducing class-level over-flagging.
- **Thresholds are calibrated, not arbitrary.** Sensitivity analysis (Figure 3.4.2) is used to understand precision/recall tradeoffs before selecting strict-noise settings.

Overall, the drift between train and test is small enough to support consistent evaluation assumptions, while the strict-noise detector remains conservative and interpretable.

3.6 Section Summary

RSITMD exhibits strong visual-language coupling with consistent caption semantics and interpretable image statistics. The proposed three-layer diagnostics provide a practical quality-control stage before retrieval model training, while controlling category-prior bias through image-dominant contradiction scoring.

Table 7: Compact multimodal reliability summary (train/test).

Metric	**Train**	**Test**
Captions	21,455	2,260
Avg words/caption	10.3	10.2
Vocabulary (raw / clean)	3,567 / 3,470	1,218 / 1,142
Captions with color cues	9,009 (42.0%)	898 (39.7%)
Captions with spatial cues	16,718 (77.9%)	1,715 (75.9%)
Caption variability (mean std)	1.75	1.68
Strict noise (HIGH / MEDIUM / LOW)	4 / 206 / 4,081	1 / 28 / 423

Table 8: Top strict-noise examples and predicted type (train).

Filename	**Category**	**Noise type**	**Score**
square_3997.tif	square	cross_modal_mismatch	0.661
port_3094.tif	port	cross_modal_mismatch	0.637
farmland_1676.tif	farmland	cross_modal_mismatch	0.627
stadium_4227.tif	stadium	cross_modal_mismatch	0.623
playground_2803.tif	playground	semantic_drift	0.618

4 Conclusion

Our comprehensive EDA reveals a robust dataset across all formats. While tabular features are mostly complete, textual data requires standardizing (lowercasing/stopword removal). Images show sufficient variety but exhibit a long-tail distribution in certain categories. The multimodal analysis confirms that text and images complement each other well.

5 References

[1] Z. Yuan, W. Zhang, K. Fu, X. Li, C. Deng, H. Wang, X. Sun. *Exploring a Fine-Grained Multiscale Method for Cross-Modal Remote Sensing Image Retrieval.* IEEE Transactions on Geoscience and Remote Sensing, 2021.